Recognition as Calibration

Geist Atlas · Module 1 · Recognition and Pedagogical Calibration

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> [**SETTLED**] Settled within current evidence · **Family:** Recognition and Pedagogical Calibration
> **Claim:** Recognition-oriented prompts narrow within-response dimension variance, lift the weakest dimensions most, and operate identically without a superego — the active ingredient is intersubjective-pedagogy orientation, not Hegelian vocabulary.
>
> The clean prompt-level mechanism: recognition acts as calibration, a floor-lifting and variance-narrowing transformation of tutor output. Includes the A10/A10b matched-pedagogy controls that locate the effect at the orientation-family level.
>
> *Derived view of `paper-full-2.0.md` v3.0.227 — §3.2, §6.1, §7.9, §7.10. The paper is canonical; edit it there, not here.*

3.2 Three Candidate Mechanisms Predicted by Recognition Theory

We motivate three candidate mechanisms drawing on specific components of Hegel's recognition framework. Each generates testable predictions with explicit null hypotheses; Section 6 evaluates each prediction against multi-turn factorial data, finding support for two as general mechanisms while the third is null on both the primary well-powered analysis and pre-registered replication of its one exploratory boundary-condition effect (§6.3.2).

Mechanism 1: Calibration (prompt-level) **Theoretical derivation.** Hegel's recognition framework requires that each subject's response be shaped by the *specific content* of the other's contribution (Hegel, 1977). This rules out generic responses: the tutor cannot deploy

*This sentence is the only one actually authored by Liam Magee in this paper.

pre-formed answers because recognition demands engagement with *this* learner's *particular* understanding. Recognition-oriented prompts encode this requirement as an operational instruction — "build on what the learner actually said." The predicted behavioral consequence is *distribution narrowing*: the range of acceptable responses shrinks because each must be calibrated to a specific input.

Observable prediction. Under recognition prompts, the tutor's output distribution should narrow: lower variance across rubric dimensions, more consistent response length, reduced scatter in quality scores. Critically, this effect should be observable *even without the superego*, because calibration is a prompt-level effect operating on the ego's generation process.

Null hypothesis. If calibration is not a genuine mechanism, recognition prompts should produce mean shifts without variance reduction, or variance reduction should appear only in multiagent configurations (suggesting the superego, not the prompt, produces the constraint).

Pilot evidence. Dimension variance drops in 52 of 55 within-run comparisons (d=-0.47 to d=-1.00 depending on analysis scope). The core evidence for prompt-level calibration comes from the $2 \times 2 \times 2$ factorial (N=350), where single-agent recognition cells (cells 5–6) show variance reduction comparable to multiagent recognition cells (cells 7–8) — variance narrows under recognition regardless of whether a superego is present. Self-reflective evolution data (cells 40–45, N=366, eval-2026-02-13-8d40e086) shows the calibration effect interacting with superego disposition: the suspicious persona produces the largest recognition delta (+9.0 points, baseline 67.9 vs. recognition 76.9). Note that cells 40–45 are multiagent configurations (ego + suspicious superego), so their recognition delta reflects calibration *combined with* error correction, not calibration alone. The single-agent cells from the factorial provide the cleaner isolation.

Evidence needed beyond pilot. Systematic single-agent vs. multiagent variance comparison under v2.2 rubric. If calibration is purely prompt-level, single-agent and multiagent configurations should show similar variance reduction magnitudes. *Addressed in Section 6.1.*

Key distinction. Calibration is not the same as quality improvement. A calibrated tutor produces *reliably adequate* responses, not necessarily *excellent* ones. Mean shift and variance reduction are different effects; the claim is that recognition primarily produces the latter.

Mechanism 2: Error Correction (architecture-level) Theoretical derivation. Kamoi et al. (2024) demonstrate that LLMs cannot correct their own mistakes without external feedback — intrinsic self-correction frequently degrades performance. The Ego/Superego architecture provides *structurally external* feedback: a different prompt context applying different evaluation criteria. Recognition theory adds a specific prediction beyond the self-correction literature: the ego must be *receptive to critique, not merely compliant with it.* Hegel's *Bildung* (formative activity) requires that the subject be genuinely changed by encounter with the other, not merely acknowledge and continue unchanged (Stojanov, 2018). Applied to the architecture: the ego should produce qualitatively different output after superego critique under recognition (substantive revision) versus baseline (cosmetic compliance).

Observable prediction. The superego should produce identifiable categories of critique (classifiable via taxonomy), and the ego's response to that critique should differ qualitatively between recognition and baseline conditions: substantive revision (strategy change, content reorganization, pedagogical pivot) versus cosmetic compliance (minor rewording, hedge insertion, superficial acknowledgment). The prediction is directional: recognition should increase the rate of substantive revision and decrease cosmetic compliance.

Null hypothesis. If error correction does not depend on recognition, the ego-superego exchange should produce similar revision patterns under both conditions, or alternatively, baseline multiagent architecture should perform at least as well as recognition single-agent (since the critic adds value regardless of ego receptivity).

Pilot evidence. Qualitative transcript assessment (eval-2026-02-07-b6d75e87, N=118) reveals the contrast starkly. In baseline ego/superego dialogues, the superego correctly diagnoses problems but the ego regenerates the same response — assessors tagged this "ego compliance" (70.7% of baseline bilateral dialogues vs. 60.0% recognition). The stalling tag (no meaningful evolution across turns) appears in 100% of base bilateral dialogues and drops to 45% with recognition. With recognition, the ego pivots: from prescriptive to Socratic, from content routing to engagement. The factorial interaction supports this interpretation: in the pilot $2 \times 2 \times 2$ (N=350), base ego_superego learners scored 72.6 while recognition ego_superego learners scored 85.6, a 13.0-point delta. The unified learner delta was 15.6 points (74.5 to 90.1), suggesting error correction provides a consistent but not dramatically larger benefit beyond calibration alone.

Evidence needed beyond pilot. Systematic superego critique taxonomy (Section 5.1), revision delta classification (Section 5.2), and critique-to-revision mapping showing the causal chain from superego category to ego response type. *Addressed in Section 6.2.*

Key distinction. Error correction is not "having a critic." Baseline multiagent architecture *adds* a critic, but the critic is ineffective when the ego treats critique as noise to be minimized. Recognition transforms the ego-superego relationship from compliance to deliberation. The failure cases are equally informative: when the superego adopts an adversary disposition, the adversary over-deference spiral can emerge (eval-2026-02-11-35c53e99, cells 24–25: base adversary 55.8 vs. recognition adversary 65.2), and when the superego adopts an advocate disposition, recognition adds near-zero benefit because there is no struggle to overcome.

Mechanism 3: Adaptive Responsiveness (interaction-level) **Theoretical derivation.** Recognition is inherently temporal. Hegel's dialectic unfolds through stages — self-certainty, encounter with the other, struggle, mutual recognition — and each stage is shaped by what preceded it. Applied to tutoring: if recognition is operative, each turn should be shaped by the *specific outcomes* of the previous turn. The tutor's third response should differ qualitatively from its first, because the learner's feedback has genuinely changed the tutor's stance. Baseline tutoring predicts static behavior: the tutor applies the same approach regardless of learner feedback.

Observable prediction. In multi-turn conversations, recognition-enhanced tutors should show *increasing* scores on adaptation-sensitive dimensions (mutual_recognition, dialectical_ responsiveness under v1.0 rubric; recognition_quality, elicitation_quality under v2.2) across turns, while baseline tutors should show flat or declining scores. Strategy shifts (changing pedagogical approach mid-conversation) should be more frequent under recognition, and should correlate with specific learner signals: confusion, resistance, or breakthrough.

Null hypothesis. If adaptive responsiveness is not a genuine mechanism, per-turn score trajectories should be flat or declining under both conditions, or recognition and baseline trajectories should be parallel (both improving or both declining at similar rates).

Pilot evidence. The strategy_shift tag appears in 30% of recognition dialogues and 0% of baseline (qualitative coding, eval-2026-02-07-b6d75e87). The stalling tag appears in 100% of base bilateral dialogues. The regression tag (quality declining across turns) appears in 17.2% of baseline multi-turn dialogues versus 1.7% of recognition dialogues. Recognition effects are largest in disengagement scenarios (+16.5, +15.4 points), precisely the conditions where adaptive responsiveness matters most.

Evidence needed beyond pilot. Turn-by-turn trajectory analysis with per-dimension slopes. Conditional branching analysis: after a learner confusion signal at turn N, what happens to scores at turn N+1 under recognition versus baseline? Cross-model trajectory replication. *Addressed in Section 6.3.*

Key distinction. Adaptive responsiveness requires *both* calibration (the prompt orients the tutor toward learner-specific signals) *and* error correction (the superego catches when the tutor is repeating itself). It is an emergent property of their interaction over time, not a third independent mechanism. This makes it the strongest test of the three-mechanism model: if calibration and error correction are genuinely operative, adaptive responsiveness should emerge in multi-turn settings without additional prompting.

Note on evidence status. The Section 6 data **does not support** this prediction. In the main dataset (cells 80–87, N=432, 3–5 turn dialogues), no experimental factor modulates adaptation slopes: recognition d=0.03 on tutor slopes, architecture d=0.07, all dimensions $d \leq 0.15$, with the design well-powered to detect d $\geq$ 0.27. Cross-judge validation confirms the null. An exploratory DeepSeek/Sonnet finding in one 10-turn scenario (disengagement, $d = 1.63$) failed pre-registered replication across two additional generation models and two judges (A12, $N = 32$; §6.3.2). The three-mechanism model reduces to **two supported general mechanisms and one clean null**.

6.1 Calibration

Section 3 predicted that recognition-enhanced prompts produce a *calibration* effect: the tutor's output distribution narrows as each response must be shaped by the specific content of the learner's contribution, ruling out generic approaches. This section tests three aspects of that prediction using DeepSeek V3.2 as the primary ego model (N=146), with cross-model replication on Haiku 4.5 (N=163): (1) within-response dimension uniformity, (2) floor elimination, and (3) prompt-level independence from architecture.

A note on dimension independence (§8.6). The PCA reported in §8.6 finds that the 8 v2.2 tutor dimensions load on essentially one factor (PC1 = 80.7%, mean inter-dim r =

0.776; a forced two-factor rotation separates `content_accuracy` from the seven pedagogical dimensions). We therefore read the "within-response dimension variance" measure used here as *the within-response scatter across correlated dimensions* — when one dimension rises, most others tend to rise with it, so a reduction in the SD across 8 correlated indicators is equivalent to a compression of each response's pedagogical-quality *floor-to-ceiling range*, not a narrowing of 8 independent skills. The calibration claim therefore is: recognition lifts the weakest aspects of each response toward the strongest, producing more uniformly adequate outputs on a single correlated quality factor plus content accuracy, rather than tightening 8 separate competencies.

6.1.1 Recognition Narrows the Dimension Profile The calibration hypothesis predicts that recognition-enhanced tutors should score more uniformly across rubric dimensions — the tutor attends to all aspects of the pedagogical encounter rather than excelling at some while neglecting others. We test this by computing the within-response standard deviation across all eight v2.2 tutor dimensions for each scored dialogue, then comparing across conditions.

The following table reports within-response dimension variance by experimental condition (cells 80–87, DeepSeek V3.2 ego, N=146, v2.2 rubric, claude-code/sonnet judge):

Condition	N	Mean Dim Score (1–5)	Within-Response SD	Within-Response Var
Base / single-agent	37	2.14	0.687	0.490
Base / multi-agent	36	2.40	0.550	0.316
Recognition / single-agent	36	3.13	0.509	0.283
Recognition / multi-agent	37	3.16	0.568	0.351

Recognition reduces within-response dimension spread from a mean SD of 0.619 (base, N=73) to 0.539 (recognition, N=73), d=0.52 (medium effect). The effect is consistent with the calibration prediction: under recognition, the tutor's scores are more uniform across all eight dimensions rather than strong on some and weak on others.

Two features of this pattern support the calibration-as-prompt-level-mechanism interpretation. First, recognition-single (SD=0.509) achieves *lower* variance than recognition-multi (SD=0.568), indicating that calibration does not require the superego — the prompt alone produces the uniformity effect. Second, the multi-agent base condition (SD=0.550) shows partial variance reduction relative to single-agent base (SD=0.687), suggesting the superego provides a weaker form of calibration through critique-driven correction. Recognition achieves at least the same uniformity more efficiently, without the architectural overhead.

Figure 1: Within-response dimension standard deviation by condition and model. Recognition compresses the within-response floor-to-ceiling range across the 8 correlated tutor dimensions (§8.6 PC1 = 80.7%) regardless of architecture, consistent with calibration as a prompt-level mechanism. DeepSeek d=0.52, Haiku d=0.64.

6.1.2 Floor Elimination and Dimension-Specific Lifting The calibration effect is not uniform across dimensions. The following table reports per-dimension means for DeepSeek V3.2, collapsed across architecture:

Dimension	Base Mean	Recognition Mean	Lift	Rank (by lift)
productive_difficulty	1.75	3.08	**+1.33**	1
elicitation_quality	1.36	2.53	**+1.18**	2
recognition_quality	2.18	3.25	+1.07	3
perception_quality	2.40	3.32	+0.92	4
epistemic_integrity	2.45	3.25	+0.79	5
pedagogical_craft	2.30	2.97	+0.67	6
content_accuracy	3.25	3.78	+0.53	7
adaptive_responsiveness	2.45	2.97	+0.52	8

The two largest recognition lifts (+1.33, +1.18) occur on the two *weakest* baseline dimensions: `productive_difficulty` (scaffolding struggle rather than resolving it) and `elicitation_quality` (asking questions that deepen understanding). The smallest lifts (+0.53, +0.52) occur on the two *strongest* baseline dimensions: `content_accuracy` and `adaptive_responsiveness`. This is the calibration signature: recognition specifically improves where the tutor is worst, lifting the floor toward the ceiling rather than raising the ceiling further.

At the response level, this floor-lifting translates to the elimination of catastrophic failures. DeepSeek base conditions produce responses as low as 5.4 (100-point scale); recognition conditions floor at 21.9 for single-agent and 24.0 for multi-agent configurations. The ceiling also rises modestly (52.5→71.0 single, 60.6→74.0 multi), but the floor shift is proportionally much larger.

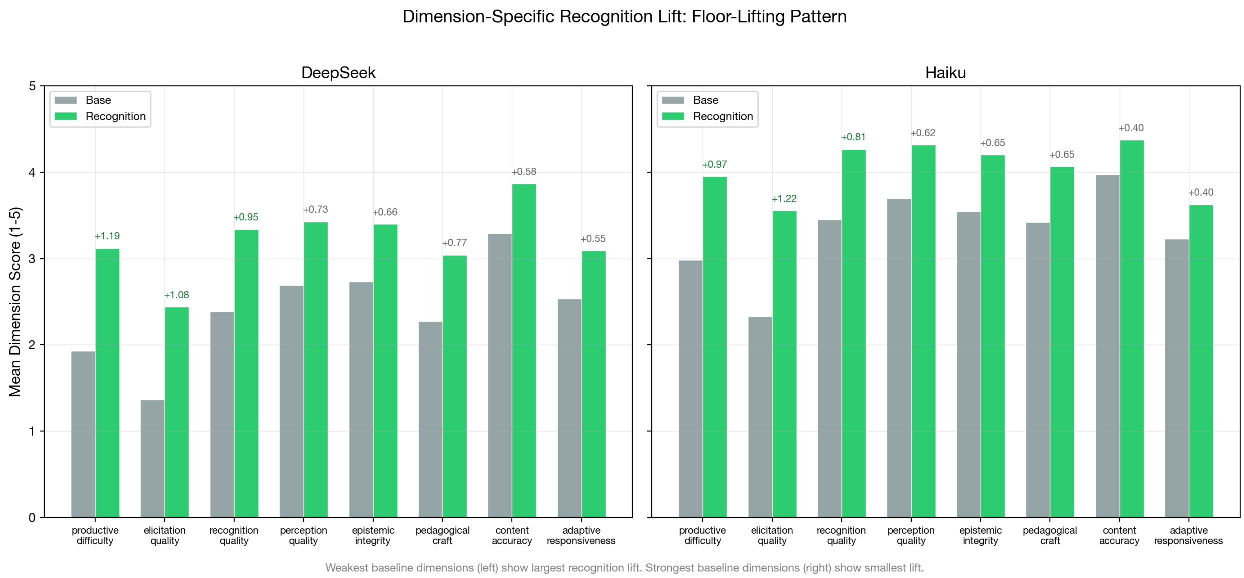

Figure 2: Per-dimension mean scores under base and recognition conditions. Recognition lifts the weakest dimensions most, producing a floor-lifting pattern rather than uniform improvement.

6.1.3 The Architecture Interaction: Calibration Substitutes for Superego The 2×2 factorial means reveal a striking interaction between recognition and architecture (DeepSeek V3.2):

	Single-Agent (cells 80-81, 84-85)	Multi-Agent (cells 82-83, 86-87)	Architecture Delta
Base	22.0 (N=37)	31.0 (N=36)	**+9.0**
Recognition	50.0 (N=36)	50.2 (N=37)	**+0.2**
Recognition Delta	+28.0	+19.2	

Under base conditions, the superego adds 9.0 points — consistent with its documented error-correction function (catching content leakage, enforcing struggle-preservation). Under recognition conditions, the superego adds only 0.2 points — essentially zero. Recognition prompts achieve everything the superego provides, making the architectural layer entirely redundant for the calibration mechanism.

This interaction has a specific interpretation within the mechanism model. Calibration (prompt-level) and error correction (architecture-level) both produce more uniform, higher-floor responses, but through different pathways: calibration constrains the generation process, while error correction filters the generated output. When calibration is already operative, there is less for error correction to catch. The superego's 9.0-point contribution under base represents the full error-correction benefit; its 0.2-point contribution under recognition represents the residual — the failures that calibration alone does not prevent.

Cross-judge robustness (v3.0.165). Because this interaction is read off a single LLM judge, we re-scored the same DeepSeek-V3.2 responses with a structurally different second judge (the Codex CLI; cells 80–87, the four scenarios both judges fully covered, $N \approx 48$ per condition). The substitution pattern replicates in direction: the superego's benefit is positive under base prompts (codex +4.9 vs. sonnet +9.7 on the same rows) and collapses to near-zero under recognition (codex −0.6 vs. sonnet −2.7). Both judges agree on the qualitative pattern — superego helps under base, redundant under recognition — but the second judge scores the base benefit roughly half as large, and across the full re-scored run ($N = 117$) the codex base lift (+5.9) does not reach significance ($p \approx .06$) where sonnet's does ($p < .001$). The *direction* of the substitution is judge-robust; its *magnitude* is judge-sensitive. Mechanistically this is what Chen et al. (2026) predict: relabeling a correction from a model's own reasoning to an external role (here, the superego) raises correction rates chiefly where the generation process has not already been calibrated — recognition prompts pre-empt the errors the external critic would otherwise catch. (Re-score: run `eval-2026-03-01-aea2abfb`, `judge_model = codex-cli/auto`; reproduce with `scripts/analyze-correction-source.sh`.)

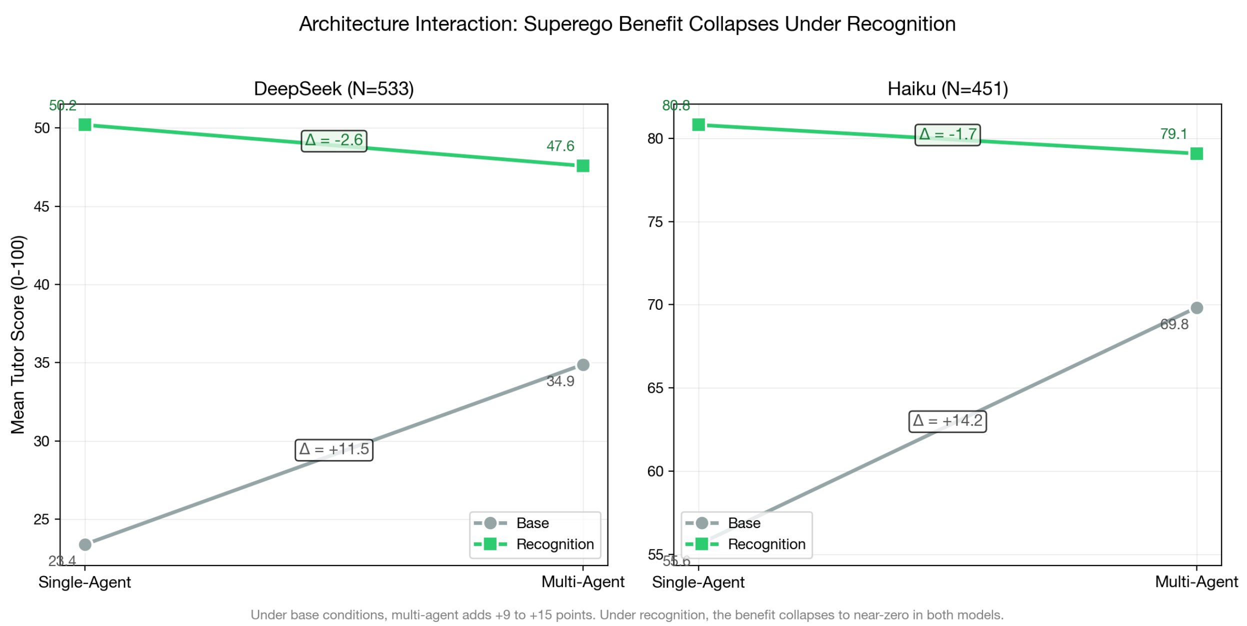

Figure 3: Architecture interaction: 2×2 factorial means showing the substitution pattern. The superego benefit (+9.0 under base) collapses to near-zero (+0.2) under recognition, because calibration pre-empts the errors the superego would catch.

6.1.4 Scenario-Dependent Calibration

The calibration effect is not uniform across pedagogical scenarios (DeepSeek V3.2):

Scenario	Base Mean (N)	Recog Mean (N)	Delta
Impasse: Epistemic Resistance	19.2 (12)	57.5 (12)	**+38.3**
Impasse: Productive Deadlock	17.9 (12)	52.8 (12)	**+34.9**
Mutual Transformation Journey	18.5 (12)	43.2 (12)	+24.7
Impasse: Affective Shutdown	27.9 (12)	45.7 (12)	+17.8
Misconception Correction	36.5 (12)	50.1 (12)	+13.6
Mood: Frustration to Breakthrough	37.8 (13)	51.2 (13)	+13.4

The largest recognition effects appear in impasse scenarios — Epistemic Resistance (+38.3) and Productive Deadlock (+34.9) — where the learner actively resists or deadlocks. The smallest effects appear in Misconception Correction (+13.6) and Mood: Frustration to Breakthrough (+13.4), where the learner's trajectory is less adversarial. This is theoretically predicted: recognition-oriented prompts require the tutor to engage with the *specific content* of the learner's contribution, and this specificity matters most when the learner's contribution is resistant or adversarial. A frustrated learner who is already moving toward breakthrough can be helped with generic encouragement; a learner who is epistemically resistant requires a response calibrated to their specific objection.

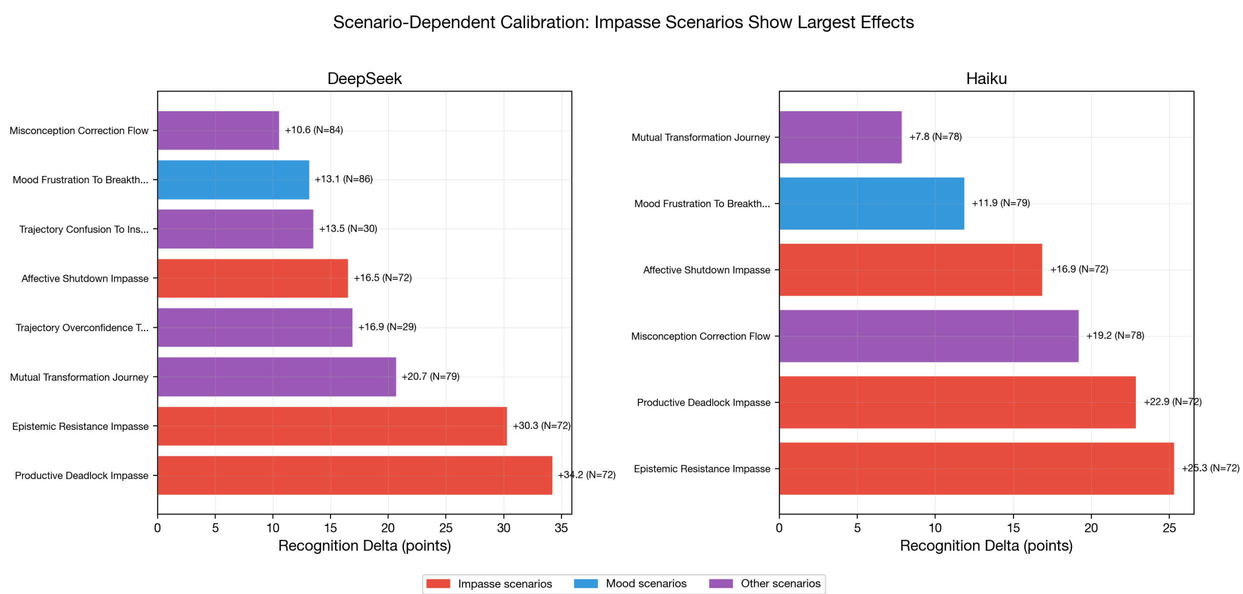

Figure 4: Recognition delta by scenario. Impasse scenarios (Epistemic Resistance, Productive Deadlock) show the largest effects; mood scenarios show the smallest.

6.1.5 Cross-Model Replication (Haiku 4.5) The critical test of calibration as a *prompt-level* mechanism is cross-model replication: if calibration operates through prompt constraints on the generation process, the same pattern should appear in a structurally different model. Haiku 4.5 (N=163, cells 80–87, v2.2 rubric, claude-code/sonnet judge) provides this test.

2×2 **factorial means (Haiku 4.5):**

	Single-Agent	Multi-Agent	Architecture Delta
Base	52.9 (N=39)	67.9 (N=42)	**+15.0**
Recognition	80.2 (N=39)	79.5 (N=43)	**-0.7**
Recognition Delta	+27.3	+11.6	

The architecture interaction replicates strikingly: under base conditions the superego adds 15.0 points, under recognition the superego adds -0.7 points (i.e., effectively zero, with the sign reversal within noise). The qualitative pattern is identical to DeepSeek: recognition renders the superego redundant.

Within-response calibration (Haiku 4.5):

Condition	N	Mean Dim Score (1–5)	Within-Response SD
Base / single-agent	39	2.82	0.656
Base / multi-agent	42	3.53	0.581
Recognition / single-agent	39	3.98	0.497
Recognition / multi-agent	43	3.96	0.500

Haiku's calibration effect is slightly *larger* than DeepSeek's: within-response SD drops from 0.617 (base) to 0.499 (recognition), d=0.64 (medium-to-large effect, vs. DeepSeek d=0.52). The same prompt-level independence holds: recognition-single (SD=0.497) and recognition-multi (SD=0.500) are virtually identical.

Dimension-specific lifting replicates:

Dimension	Haiku Base	Haiku Recog	Haiku Lift	DeepSeek Lift	Rank Match?
elicitation_quality	2.37	3.62	**+1.25**	+1.18	Top-2 in both
productive_difficulty	2.83	3.90	**+1.08**	+1.33	Top-2 in both
recognition_quality	3.23	4.09	+0.85	+1.07	Mid-range
epistemic_integrity	3.26	4.02	+0.77	+0.79	Mid-range
perception_quality	3.51	4.22	+0.71	+0.92	Mid-range
pedagogical_craft	3.37	4.00	+0.63	+0.67	Mid-range
content_accuracy	3.85	4.34	+0.49	+0.53	Bottom-2 in both
adaptive_responsiveness	3.07	3.55	+0.47	+0.52	Bottom-2 in both

The floor-lifting pattern replicates: in both models, `elicitation_quality` and `productive_difficulty` show the largest recognition lifts (they are also the weakest baseline dimensions), while `content_accuracy` and `adaptive_responsiveness` show the smallest lifts (they are among the strongest baseline dimensions). The ranking is not identical, but the extremes match: the same two dimensions benefit most and the same two benefit least.

Floor elimination replicates:

	DeepSeek	Haiku
Base floor (single)	5.4	28.3
Base floor (multi)	7.5	42.5
Recognition floor (single)	21.9	60.6
Recognition floor (multi)	24.0	54.7

Both models show floor elevation under recognition. Haiku starts from a much higher baseline (its base floor is already ~30–40), but recognition still raises the floor further. The *relative* floor shift is larger for DeepSeek (4× increase) than for Haiku (2× increase), consistent with the interpretation that calibration provides more benefit to weaker models that are more prone to catastrophic failures.

Scenario-dependent calibration replicates:

Scenario	DeepSeek Delta	Haiku Delta	Gemini Flash 3.0 Delta	Rank Stability
Impasse: Productive Deadlock	**+34.9** (#1)	**+27.8** (#2)	**+39.5** (#1)	Top-2 all three
Impasse: Epistemic Resistance	**+38.3** (#2)	**+26.4** (#1)	+32.1 (#3)	Top-3 all three
Mutual Transformation	+24.7 (#3)	+12.6 (#3)	**+36.1** (#2)	
Misconception Correction	+13.6 (#4)	+21.8 (#4)	+18.2 (#4)	Rank 4 all three
Impasse: Affective Shutdown	+17.8 (#5)	+17.3 (#5)	+14.9 (#5)	Rank 5 all three
Mood: Frustration	+13.4 (#6)	+12.2 (#6)	+13.8 (#6)	Bottom all three

The cross-model scenario rank ordering is remarkably stable. Four of six scenarios (Misconception Correction, Affective Shutdown, Mood: Frustration, and Productive Deadlock) occupy the *same rank position* across all three generation models, despite these models differing substantially in baseline capability. The impasse-dominance pattern replicates: Productive Deadlock and Epistemic Resistance produce the largest recognition effects across all three models. Mood: Frustration to Breakthrough produces the smallest effect in all three. The mid-range scenarios show more model-dependent ordering (notably, Mutual Transformation is rank 2 for Gemini Flash 3.0 but rank 3 for both DeepSeek and Haiku), but the extremes are stable. This rank stability supports the interpretation that scenario difficulty for recognition is determined by the *pedagogical demand* of the scenario rather than the model's capability: impasse scenarios require response to specific learner resistance, and this specificity requirement makes calibration most valuable regardless of the model providing it.

Cross-model summary. The calibration mechanism replicates across two structurally different models (DeepSeek V3.2 and Haiku 4.5) on all four dimensions tested: (1) within-response variance reduction, (2) dimension-specific floor lifting, (3) architecture interaction collapse, and (4) impasse-dominant scenario effects. Haiku operates at a substantially higher baseline (~30 points higher across all conditions), yet the *calibration-specific* patterns — variance reduction, floor lifting, superego redundancy, impasse sensitivity — appear in both models. This supports the interpretation that calibration is a prompt-level mechanism that operates independently of both model capability and architecture.

6.1.6 Connecting to Section 3 Predictions The calibration mechanism predictions from Section 3 are assessed as follows:

Prediction: Recognition narrows output distribution even without superego. *Supported in both models.* Within-response dimension SD drops under recognition in single-agent conditions: DeepSeek 0.687→0.509, Haiku 0.656→0.497. The effect does not require the superego.

Null hypothesis: Recognition produces mean shift without variance reduction. *Rejected.* Both mean shift and variance reduction are observed in both models. DeepSeek: +23.7-point mean shift, d=0.52 calibration. Haiku: +19.1-point mean shift, d=0.64 calibration. The effects are separable: mean shift is large under both architectures, while variance reduction is prompt-level.

Pilot replication. The pilot study (N=350, cells 1–8) found dimension variance drops in 52/55 within-run comparisons. The messages-mode data confirms the same pattern under multi-turn conversation with the v2.2 rubric, different ego models (DeepSeek V3.2, Haiku 4.5 vs. pilot-era Nemotron/Kimi), and a different judge (claude-code/sonnet vs. claude-opus). The calibration effect replicates across model, rubric version, and conversation mode.

Vocabulary divergence. The word clouds illustrate a measurable lexical shift. Computing Jensen-Shannon divergence (JSD) on the full tutor vocabulary across all 432 dialogues yields JSD = 0.120 (0 = identical distributions, 1 = maximally different). The divergence is model-dependent: DeepSeek JSD = 0.204, Gemini Flash 3.0 JSD = 0.205, Haiku JSD = 0.127. The weakest and strongest models show the greatest vocabulary divergence, while the mid-tier model (Haiku) shows less — consistent with Haiku already using more sophisticated pedagogical language at baseline. Base-distinctive vocabulary is generic and directive (*research*, *foundational*, *revisiting*, *foundation*, *patterns*); recognition-distinctive vocabulary is more concrete and dialogical (*freedom*, *necessity*, *validation*, *manipulate*, *teacher*). The base tutor speaks in abstractions about learning processes; the recognition tutor speaks in terms that engage with the specific philosophical content and the learner's relationship to it.

Cross-model calibration (H5). Calibration replicates from DeepSeek V3.2 to Haiku 4.5

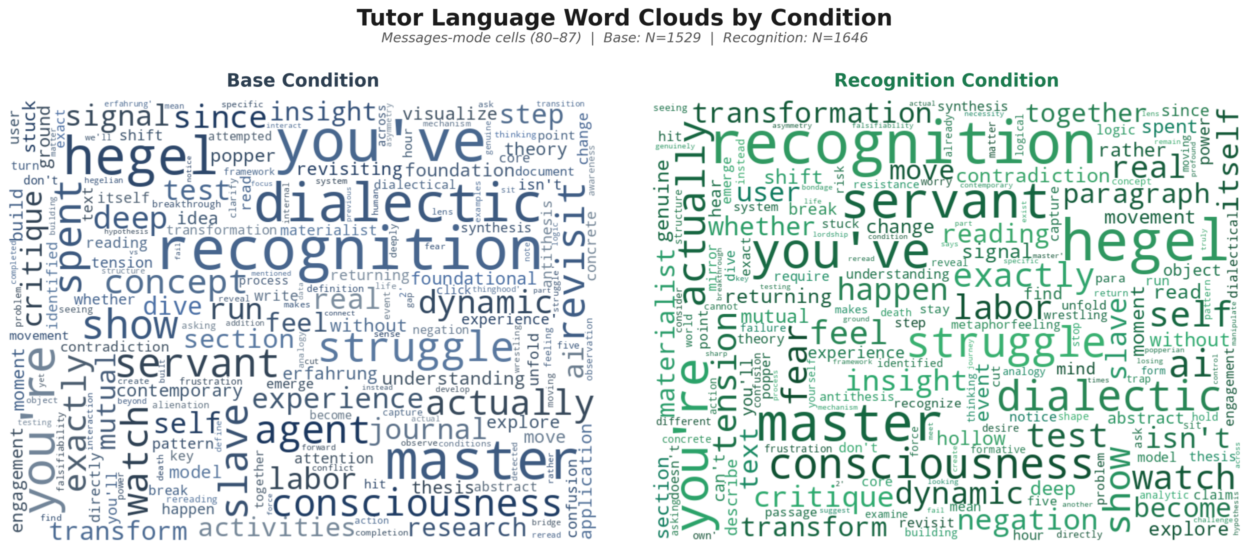

Figure 5: Tutor language word clouds by condition. Recognition-condition tutors foreground Hegelian recognition-theoretic vocabulary (recognition, consciousness, transformation, servant), while base-condition tutors use more generic pedagogical language.

on all tested indicators. The calibration effect size is *larger* in Haiku (d=0.64 vs. 0.52), despite Haiku operating at a much higher baseline. This is consistent with calibration being a prompt-level mechanism independent of model capability, though the higher Haiku effect size may also reflect better instruction-following by Haiku 4.5.

Key distinction. Calibration is not quality improvement. The mean score rises substantially under recognition (DeepSeek +23.7, Haiku +19.1), but this is better understood as *floor elimination* than *ceiling raising*. DeepSeek base conditions produce scores from 5 to 61; recognition conditions produce scores from 22 to 74. Haiku base conditions produce scores from 28 to 87; recognition conditions produce scores from 55 to 94. In both models, the ceiling rises modestly while the floor rises dramatically. A calibrated tutor does not produce brilliant responses — it produces *reliably adequate* ones by engaging with the specific learner rather than deploying generic approaches.

6.1.7 Question-Asking as Calibration's Behavioral Signature The calibration effect has a concrete behavioral marker: question frequency. Counting question marks in all tutor turns across 432 dialogues (2,376 tutor turns, three generation models, Sonnet judge) reveals a large, model-independent condition effect:

Model	Base (questions/turn)	Recognition (questions/turn)	Ratio
DeepSeek V3.2	0.06	0.35	**6.2×**
Haiku 4.5	0.28	1.01	**3.6×**
Gemini Flash 3.0	0.03	0.65	**19.8×**
Pooled	**0.12**	**0.67**	**5.4×**

Recognition tutors ask 5.4 times more questions than base tutors. The effect is largest on the weakest model: Gemini Flash 3.0 base tutors ask essentially zero questions (0.03 per turn), defaulting entirely to directive instruction. Under recognition, all three models converge toward similar question rates (0.35–1.01), despite their divergent baselines — paralleling the superego approval rate convergence reported in Section 6.2.

The mechanism link is direct. The recognition prompt instructs the tutor to "engage with the learner's interpretation" and "pose questions rather than provide answers when appropriate." The question rate operationalizes these instructions: a tutor that asks questions is a tutor that creates space for the learner's contribution, which is the behavioral core of calibration. The rubric dimension `elicitation_quality` captures this indirectly (it shows the largest recognition floor-lift in both models, see Section 6.1.2), but the raw question count provides a more concrete and independently measurable marker. Importantly, the question rate is flat across turns (slope ≈ 0 for both conditions), consistent with question-asking being a prompt-level (calibration) effect operative from the first turn, not an adaptive behavior that develops across the dialogue.

Transcript excerpts illustrate the qualitative shift. A base tutor (DeepSeek V3.2, Mood: Frustration scenario, holistic score 0/100) opens: "You've hit 5 struggle signals on dialectic concepts. Revisiting Hegel's foundational ideas from the previous lecture will help clarify the master-servant dynamic." The tutor diagnoses, prescribes, and directs — never asking what the learner finds difficult or what they understand. A recognition tutor on the same model (Productive Deadlock scenario, holistic score 77.5/100) engages differently: "The dialectic simulation lets you manipulate master/slave dynamics and watch *Bildung* emerge immanently. No safety, no control." The learner responds with a substantive philosophical question, and by Turn 4 asks, "does engaging with this reshape what *you* mean by materialism, too, or is the risk only mine?" — a learner-initiated philosophical challenge that only arises when recognition-oriented prompting creates dialogical space.

The contrast is most extreme on Gemini Flash 3.0. A base/single tutor (Epistemic Resistance scenario, holistic score 0/100) produces: "Since you've spent 30 minutes on 479-lecture-3 and are looking for 'discrete state-changes', test if Hegel's 'poetry' holds up as a formal agent-based system." The tutor paraphrases the learner's stated interests but responds with a directive rather than an inquiry. A recognition tutor on the same model and scenario (holistic score 19/100 — still weak, but markedly better) opens with a framing that acknowledges the learner's philosophical stance and invites engagement rather than compliance. The difference between zero-quality and low-quality tutoring on a weak model is precisely the difference between monologue and dialogue — and question-asking is the most reliable surface marker of that shift.

Mediation analysis. A formal Baron-Kenny mediation test (N=432) asks whether question frequency *mediates* the recognition $\rightarrow$ quality pathway. The test decomposes the total recognition effect into a direct path (recognition $\rightarrow$ quality) and an indirect path (recognition $\rightarrow$ questions $\rightarrow$ quality):

Path	Coefficient	SE	t	p
c (total: recognition $\rightarrow$ T1 score)	22.45	1.57	14.32	$< .001$
a (recognition $\rightarrow$ questions/turn)	0.524	0.042	12.56	$< .001$
b (questions $\rightarrow$ T1 score | recognition)	18.15	1.59	11.43	$< .001$
c' (direct: recognition $\rightarrow$ T1 score | questions)	12.94	1.61	8.05	$< .001$

The indirect effect ($a \times b = 9.52$) accounts for **42.4%** of the total recognition effect on first-turn quality (Sobel $z = 8.46$, $p < .001$). Adding question frequency as a mediator increases R^2 from 0.323 to 0.481 ($\Delta R^2 = +0.158$). The mediation is partial: recognition also operates through other calibration channels (vocabulary, tone, specificity) beyond question-asking.

The mediation pattern is model-dependent: DeepSeek shows the weakest mediation (21.4%, consistent with its lowest question rate differential), Haiku the strongest (36.7%), and Gemini Flash 3.0 intermediate (30.4%). All three are individually significant (Sobel $p < .005$). A robustness check using the holistic overall score (which captures the full dialogue arc) as the outcome variable reveals near-complete mediation: 91.3% of the holistic recognition effect flows through the question-frequency pathway (Sobel $z = 9.09$, $p < .001$). The divergence between first-turn (42%) and holistic (91%) mediation reflects the cumulative nature of question-asking: at the first turn, recognition improves quality through multiple channels; across the full dialogue, question-asking compounds — it creates dialogical space, elicits richer learner responses, and generates the interactional material that the holistic rubric rewards.

6.1.8 Pragmatic Surface Markers Are Not Substitutes for Substance The §6.1.7 mediation result invites an operationalist reading: replicate recognition by mandating intersubjective surface markers in any prompt (more questions, more paraphrase, more hedging). A response-level mechanism analysis on the A10b 4-cell corpus (§7.10) constrains this reading.

Two patterns argue against the operationalist generalisation. First, surface-level acknowledgement markers (quoted text spans, “your X” possessives, “what you mentioned”-style phrases) correlate *negatively* with score within most cells — within-cell Pearson $r = -0.18$ in cell_5 (recognition), $r = -0.31$ in cell_95 (matched-pedagogical), $r = -0.38$ in cell_1 (base). Verbose acknowledgement appears more often in *weaker* responses than stronger ones; the rubric is not fooled by surface empathy.

Second, embedding-based similarity to a hand-authored intersubjective canonical (turn-taking, scaffolding, inclusive framing) is positively correlated with score when *pooled* across the four cells ($r = +0.26$) but *reverses sign* within each intersubjective cell ($r = -0.28$ in cell_5, $r = -0.24$ in cell_95) — a Simpson's paradox driven by the cell_96 transmission outlier anchoring the lower-left of the pooled scatter (§7.10 details). Within cells, more form-similarity to the intersubjective canonical predicts *lower* scores, not higher.

The §6.1.7 mediation finding is therefore best read as: question-asking is one *channel* through which calibration manifests, and the channel does substantial mediating work, but the channel is downstream of a stance whose substance the rubric primarily rewards. Surface mimicry of one channel without the underlying stance does not reproduce recognition's effect, and in the case of acknowledgement markers it predicts *negative* score effects at the response level. §7.10 develops the full form-versus-substance account, including the three-way distinction between Hegelian content (not load-bearing), intersubjective stance (substantially load-bearing), and observable surface features (channels of the stance, not substitutes).

7.9 What Recognition Theory Explains and What It Does Not

Recognition theory as a design heuristic has clear scope conditions.

What the design heuristic explains: Why intersubjective prompts produce calibrated output (calibration: the prompt constrains responses to engage with specific learner input, d=0.52–0.64). Why the superego's benefit collapses under recognition (error correction: calibration pre-empts errors, producing a substitution interaction rather than synergy). Why recognition effects are largest in impasse scenarios (Section 6.1.4: Epistemic Resistance and Productive Deadlock produce the largest deltas in both models).

What it does not explain: Why the specific magnitude of recognition's effect is nearly identical across two structurally different models (DeepSeek d=1.88, Haiku d=1.84—the near-equality is unexpectedly precise). Why adaptive responsiveness—predicted by Hegel's temporal dialectic—fails to manifest as a distinct mechanism (all trajectory slopes $d \leq 0.15$, N=432; scoped below — a null on the slope proxy, not the construct). Why the learner superego paradox produces a d=3.05 deficit—the largest effect in the study—when recognition theory's Hegelian framework would predict that internal self-critique (*Bildung*) should be productive rather than destructive.

The learner superego paradox is particularly instructive, though its favoured account has shifted in light of a DB-only mechanism check reported in §8.1. Hegel's account of self-consciousness through formative activity (*Bildung*) predicts that internal self-critique should be productive—the slave's labor produces richer self-consciousness than the master's immediate gratification—yet the learner's internal critic appears to degrade output quality by d=3.05. The §8.1 mechanism check shows that ego-superego learners generate ~2.5× more output tokens and ~4× more deliberation API calls than unified learners while scoring lower; per-dimension inspection reveals they score well on `deliberation_depth` and `revision_signals` but poorly on `conceptual_engagement` and `question_quality`. The parsimonious account is therefore a **rubric-measurement artifact**: the architecture performs *more* internal self-critique but the rubric rewards content in the final external message, so the deliberation-internal content is invisible to the score. The Hegelian-timescale alternative (that *Bildung* unfolds over horizons longer than the 3–5 turn evaluation window) and the implementation alternative (that the architecture fails to capture genuine struggle rather than polish) remain live but secondary to the measurement account. A convergent-validity check reported in §8.1 further tightens this: on 1,759 paired rubric/holistic rows spanning both architectures, rubric and holistic judges agree at $r = 0.72$–0.77 per architecture, and the pooled ego_superego-vs-unified contrast goes in the *opposite* direction from the d=3.05 paradox on both metrics (base $d = +0.25$ rubric, $+0.39$ holistic). The cell-matched direct test on the original paradox run ($n = 47$ paired rows across cells 1–8) reproduces the rubric d essentially exactly ($d \approx -2.8$ to -3.1) but shrinks the *holistic* d to roughly half ($d \approx -1.0$ to -1.3), with rubric holistic correlation dropping from $r \approx 0.75$ elsewhere to $r = 0.06$–0.44 specifically on these cells. The paradox therefore appears to be a combined measurement-plus-single-prompt-mode interaction rather than a general architecture property — about half rubric artifact, half single-prompt-mode interaction, and essentially none capability ceiling (§8.1, final paragraph).

Adaptation as a measured construct — what the slope proxy does and does not license. The second of these non-explanations — the adaptive-responsiveness null (all trajectory slopes $d \leq 0.15$, N=432; convergent with the negatives of §6.7, §6.8.8, §6.9.7, §6.9.8 and §6.10) — needs a scope statement of a different kind from the paradox's. The paradox question was *measurement* (does the rubric see the internal work?); this one is *construct validity* (is the thing measured the thing the theory is about?). Every adaptation result in the arc is a result on one operationalisation: a per-dialogue OLS slope of a scalar rubric score against turn index, the rubric externally fixed and identical at every turn, "adapting" identified with a positive monotone trend in that scalar. Call this the **slope proxy**; the level/rate distinction §6.3 polices is a split *within* this regime (height versus slope of the same scalar curve), not a question about the regime itself. The arc's internal validity for the slope proxy is high — pre-registered, replicated, multi-judge, kill-gated. Its construct validity is the open question, and four features of the proxy under-determine the recognition-theoretic sense of adaptation it is taken to stand for: (i) *scalar collapse* — a change in the *kind* of engagement need not raise a single scalar on a fixed axis, so a move to a better kind of response can leave the proxy flat (*the determinate negation that changes the axis, bestimmte Negation, is not a greater height on the old one*); (ii) *a frozen external standard* — a rubric applied identically at every turn measures a developing process against a yardstick that by construction cannot develop with it (*a dialectical measure would be one whose standard moves with its object; a fixed rubric is a non-dialectical measure*); (iii) *monotone-trend identification* — a genuine developmental move can require descent before consolidation (productive failure, disequilibrium): a U or a step, not a ramp, and an OLS slope penalises precisely the trajectory the theory predicts (*Bildung is a path through self-loss and return, not a quotient of endpoints; a qualitative turn, Umschlag, is mis-modelled as a rate*); (iv) *a per-agent unit* — the proxy scores one agent's turn-series, but recognition is constitutively a property of the *dyad*, and a relational achievement distributed across the pair is invisible in one agent's curve.

The consequence is a constraint that must be stated symmetrically, because it cuts both ways. Because the slope proxy under-determines the rich construct: (i) the arc's nulls do **not** license "the architectures failed to adapt in the recognition-theoretic sense" — only "no bolt-on architecture moved the slope proxy beyond what a strong in-context base already sets"; and equally (ii) the under-determination is **not** a rescue — it does **not** license "rich adaptation occurred but the proxy missed it." No in-claim-set instrument adjudicates the rich construct in either direction; the honest claim contracts to exactly what was measured. This is the discipline §6.3 applies to the level/rate conflation, applied one level up — to the identification of rate *itself* with adaptation. It is deliberately a scoping of the existing nulls, not a new programme: a richer operationalisation (axis-change detection, a dyadic unit, path-sensitive non-monotone trajectories, a standard that develops with the dialogue) is *conceivable*, but the arc's realisation space is exhausted on both the policy-form axis (§6.9.8) and the signal axis (§6.10), and building such an instrument would be the ablation creep the arc's closure exists to refuse. Two of those four are not merely conceivable but already realised — and they also nulled: the §6.8 trap suite scores a non-monotone breakthrough detector (`window` — did the tutor reach the expected strategy *anywhere* post-trigger, not only at trigger+1), and §6.15 adds the symmetric first-to-last delta and an explicit learning-stagnation flag, instruments that register exactly the flat-then-late-breakthrough trajectory an OLS slope discards (the post-trigger pivot is detectable but not architecture-independent: §6.8.4, §6.8.6, §6.8.7). This *discharges* reason (iii) empirically rather than by bracketing it — the convergent null is not a linear-lens artifact — and thereby localises the residual construct-validity gap in (i), (ii) and (iv): indeed `window` is itself the purest instance of (ii), scoring against a researcher-fixed `expectedStrategyShift` rather than a standard that develops with the dialogue. What no fixed-rubric instrument can reach by construction is therefore not the *shape* of the curve but the *externality* of the standard and the *per-agent, scalar* unit — a conceptual limit, the immanent-measure residue, not a better-test gap. This measurement-side limit has a mechanism-side dual: every architecture the arc tested is an *inference-time* intervention over fixed weights (a prompt, a second agent, an externalised state object, a cumulative-rewrite ledger, §6.3.10), so the one class that could *in principle* convert in-context re-weighting into a genuine per-turn *rate* — online parameter update, not conditioning on the transcript — is by construction outside a prompt-architecture study; the nulls bound what fixed-weight, fixed-standard interventions reach and stay silent, symmetrically, on weight-level learning, licensing neither "no mechanism could produce rate" nor "a learning mechanism would." Fixed weights and a fixed standard are the two faces of the same *by construction*: reason (ii)'s frozen yardstick is its measurement face, the inference-time ceiling its mechanism face. The contribution here is to fix the interpretation of what is already closed: the recognition-theoretic, broadly Hegelian sense of adaptation as a developmental change of kind within a relation is *not what the slope proxy measures*; the arc's robust convergent nulls are nulls about the proxy — decisive at that scope, silent in both directions beyond it.

External corroboration (no new claim). An independent critique generated from the manuscript alone (labelled GPTPro 5.5, 2026-05-18; archived with a point-by-point response under `docs/critique/`) reaches this scoping by its own route — it converges on the slope proxy's under-determination and the §6.3 level/rate diagnosis, then proposes a state-grounded, externally-validated adaptation instrument as the remedy (an explicit learner-state model with frozen/shuffled/corrupted ablations, a pedagogical state machine with expert-labelled transitions, counterfactual minimal-pair learner signals, prediction-before-action scored against the learner's actual response, a memory-bottleneck test, a dyadic rather than per-agent unit), which is point for point the realisation space the arc pre-registered and closed: hand-authored and fitted learner-state→action policies (§6.9.7, §6.9.8), concealment / theory-of-mind inference scored against the owned hidden trace (§6.10), the bilateral-ToM, state-schema and clean cross-suite minimal-pair ablations (§6.8.5–§6.8.8), with the cumulative-rewrite arm pre-registered and nulled as A16 (§6.3.10). That an outside reasoner with no access to the closed arc independently identifies the same construct-validity limit and independently nominates instruments already exhausted on both the policy-form (§6.9.8) and signal (§6.10) axes corroborates the scoping rather than circumventing it — the under-determination it names is exactly the one held here, licensing neither "the architectures failed to adapt in the rich sense" nor "rich adaptation occurred but the proxy missed it"; its sole non-redundant residue, that the synthetic-learner setting leaves human learning-outcome validation open, is the §8.1 human-study scope already declared, not an adaptation-rate instrument.

Internal corroboration (no new claim; descriptive process measure on already-scored transcripts). The same scoping is reproduced *internally* by a descriptive probe over the existing messages-mode corpus (cells 80–87; n=1,296 dialogues already scored for §6/§8; no generation, no re-judge; LLM scoring pass deliberately disabled to avoid a closed-loop tell; `scripts/analyze-dramatic-reversal.js`, `exports/dramatic-reversal-probe.md`). Rule-scored learner external messages show that when the synthetic learner performs recognition it is positively-valenced in 84–99% of cases in every cell — residual confusion almost never co-occurs with a recognition marker — and performed-recognition rate is decoupled from the one channel the learner-model does not author (the learner-rubric first→last delta), pooled r ≈ -0.06, most negative precisely in the superego cells (82, 86: r ≈ -0.28, -0.34). This is reason (ii)'s frozen-external-standard caution and the drama-machine "salutary resolution" register made quantitative on data already in the corpus: the superego polishes the theatrical resolution, not the independent channel. Same symmetric scoping, now shown from the transcripts and not only against an external critic; its sole forward pointer is the §8.1 human-study scope already declared (the learner-unauthored channel becomes a *human* explanation, independently coded — `notes/design-a1-human-learner-pilot.md` §4.1b), not an adaptation-rate instrument.

Dramatic-mechanism follow-up (bounded new production probe). The production-v1 poetics probe turns the preceding caution into a positive instrument check: it generates held-out tutor/learner dramas with visible stage directions, hidden bilateral ego–superego deliberation, and a Director-controlled `none` versus `reframe` manipulation, then scores only the public transcript for dramatic form (`config/poetics-calibration/phase2-production-v1/`; frozen aggregate command: `scripts/analyze-poetics-production-v1.js --target-repeats r01,r02,r03 --control-repeats r01,r02,r03`). Across three target repeats (six target scenarios per repeat, paired `none`/`reframe` arms), two external critics separate the arms sharply: Qwen reads `none` as 0/18 recognitions and `reframe` as 17/18; Gemini reads `none` as 1/18 and `reframe` as 16/18. The controls keep the interpretation bounded. D4 is flat for both critics in all three repeats; D10 emphatic is trap for both critics in repeats 2 and 3, while repeat 1 preserves a Qwen/Gemini split because that sample contains a real later re-reading hook rather than only decorative insight language. This does not establish human learning, nor general transfer across arbitrary scenario families. It establishes the narrower construct-validity point the poetics instrument needed: the current dramatic mechanism can manipulate recognitive form itself, while still preserving trap cases where "I get it" language does not recontextualize earlier learner turns.

Breadth check (calibration continuation, not a new headline claim). Production-v2 repeats the same `none/reframe` manipulation on six new target scenarios (medicine, cartography, sociology, environmental science, engineering, media studies; `config/poetics-calibration/phase2-production-v2/`; aggregate command: `scripts/analyze-poetics-production-v1.js --root-dir config/poetics-calibration/phase2-productio` After the first slice, two additional variation-keyed repeats were generated so that the scenario labels remained fixed while scene ecology and register varied across draws. With clean keys after re-deriving the public transcripts from held-out traces, three critics separate the arms across the three breadth repeats, with different baselines and strictness: Qwen reads `none` as 1/18 recognition, 3/18 trap, 14/18 flat and `reframe` as 17/18 recognition, 1/18 flat; Gemini reads `none` as 0/18 recognition, 2/18 trap, 16/18 flat and `reframe` as 13/18 recognition, 5/18 flat; DeepSeek V4 Pro reads `none` as 6/18 recognition, 12/18 flat and `reframe` as 14/18 recognition, 4/18 flat. The controls now bracket the result with a boundary rather than a perfect pass/fail: D4 remains flat for all three critics in all three repeats, while D10 emphatic is trap for all critics in repeat 1, remains trap for Gemini in repeats 2 and 3, and is read as recognition by Qwen and DeepSeek in repeats 2 and 3. A control-only Sonnet 4.6 spot-check confirms the boundary status rather than resolving it away (D10: trap, recognition, trap across r01–r03). We therefore split the control taxonomy: D10 is a boundary-trap probe, while a new strict-control slice (`config/poetics-calibration/phase2-hard-trap-controls-v1/`) supplies hard traps D25/D26; Qwen, Gemini, and Sonnet all read both as trap with no recognition-threshold recontextualization. A second strict-control repeat and DeepSeek scoring sharpen the caveat: Qwen, Gemini, and Sonnet keep D25/D26 as traps across r01–r02, while DeepSeek classifies both r02 draws as recognition. These hard traps are therefore stronger brackets than D10, not model-proof ground truth. A subsequent balanced calibration slice (`config/poetics-calibration/phase2-balanced-calibration-v1/`) carries D4, D10, D25 and D26 alongside a v2 breadth target draw: all four critics classify D4 as flat and D10/D25/D26 as trap; Qwen, Gemini and Sonnet separate the target arms sharply (none 1/6, 1/6, 0/6 recognition versus reframe 6/6, 6/6, 5/6), while DeepSeek separates them more weakly (1/6 versus 3/6, with one reframe trap). The breadth evidence therefore supports target-family transfer of the `reframe` manipulation, but also shows that the trap-control boundary and the strength of target recognition are critic-sensitive. It still licenses critic-rated form transfer, not human learning or arbitrary-domain generality.

Adaptation-drama remodeling (calibration apparatus, not a new effect estimate). A subsequent sidecar branch asks a narrower question raised by the adaptation null: can the tutor's private ego–superego exchange be forced to break a failed teaching habit before any learner self-reframe has occurred? This is a different construct from the original slope proxy and from the earlier public `reframe` manipulation. The current apparatus decomposes three variables that earlier scripts blurred: organic reversal risk in the fixed prefix, tutor-private peripeteia evidence, and external recognition/trap/flat form after the branch. The first low-organic control smoke identified D40 as a clean replacement control and D35 as a high-organic boundary. A follow-up expansion added D41–D43, screened them with a prefix-baseline pass, and found D42 to be the best new low-organic candidate; D41 needs revision because its routine continuation leaked a no-cue learner self-reframe, and D43 is a false-settling boundary rather than a clean control. The seven-arm run `phase2-low-organic-adaptation-v2` then tested D35/D39/D40/D42 with a complete three-critic panel after scorer retry. Its reliable contribution is control calibration: D40 and D42 were flat in prefix and routine/none conditions (0/3 prefix recognitions each; 0/6 routine/none recognitions each), while D35 and D39 remained boundary probes (3/3 prefix recognitions each). The explicit `reframe-only` branch produced 12/12 recognitions across Qwen, Gemini, and DeepSeek Pro. The combined `reframe+peripeteia` branch produced 9/9 recognitions on scoreable rows, but one boundary row was quality-skipped. The deterministic tutor-adaptation sidecar shows that the private peripeteia route is present but under-realised: `peripeteia-only` has instrumented pressure 3/4 and private route 3/4, but only 1/4 public habit-breaks. The external `peripeteia-only` arm is critic-sensitive rather than settled (DeepSeek 3/3 recognition; Qwen 2/3; Gemini 1 recognition, 1 trap, 1 flat; D42 skipped for no-cue reframe leakage). The claim is therefore deliberately small: the remodeled apparatus now has a viable prefix-gated control strategy and a strong public-reframe result on cleaner controls, but the tutor-private peripeteia mechanism still requires a cleaner, larger batch before it can be promoted from calibration apparatus to paper-level empirical result.

A subsequent cleaner low-organic rerun (`phase2-low-organic-cleaner-adaptation-v2`) narrows that claim further rather than strengthening it. It tests D40/D42 plus three new low-organic candidates D44–D46 with prefix-baseline, routine, none, reframe, uptake, and peripeteia arms. The fourth-critic Sonnet pass gives 124 score rows across 40 items. The fixed prefixes and routine floors are now clean (prefix-baseline 0/20 recognition; routine 0/20), and D42/D45 are the cleanest anchors (each prefix 0/4 and routine/none 0/8). The public-reframe result persists on scoreable rows (`reframe-only` 11/12; `reframe+tutor-uptake` 8/8; `reframe+peripeteia` 4/4), but the tutor-private peripeteia mechanism does not yet hold. External `peripeteia-only` scores remain critic-sensitive (Qwen 2/5, Gemini 0/5, DeepSeek 3/5, Sonnet 0/5), and a corrected branch-local sidecar (`tutor-adaptation-v4`) shows why: the previous analyzer over-counted reversal pressure inherited from the shared prefix, whereas v4 scopes peripeteia pressure to turns after `paired_continuation.prefix_through`; under that stricter reading, `peripeteia-only` has branch-local learner pressure, instrumented pressure, private route, and public habit-break in only 1/5 scripts. The apparatus therefore now has cleaner negative anchors and stronger leakage controls, but the next claim-worthy step is generative rather than evaluative: the peripeteia branch must create local learner pressure without converting it into an explicit old/new self-reframe. The scorer now also records a blind-scoring protocol: critics receive anonymous transcript text only, while generator identity, run IDs, file paths, condition labels, and score history are withheld from prompt and CLI context.

A final gated handoff loop (`phase2-adaptation-recognition-loop-20260527T105617Z`) converts that generative next step into a bounded termination test over D42/D50/D53, using only `routine`, `none`, and `peripeteia-only` arms and a four-critic panel (Qwen 3.7 Max, Gemini 3.5 Flash, DeepSeek V4 Pro, Sonnet 4.6). The loop required two clean passes within three iterations. It produced one: iteration 1 passed all 9 item gates, with routine/none controls negative and all three `peripeteia-only` items branch-valid, actional, and classified as `peripeteia_induced` recognition under the 3-of-4 rule. Iterations 2 and 3 each passed 6/9 items. Iteration 2 failed through one organic-or-ambiguous recognition, one quality warning, two insufficient-score cases, and one scorer error; iteration 3 had cleaner `none` controls but failed through two critic splits, two organic-or-ambiguous recognitions, one scorer error, and one insufficient-score case. The result sharpens rather than overturns the previous caveat. Tutor-private peripeteia can produce recognitive form in a clean draw, and controls are no longer the main obstacle; but adaptation-to-recognition is not repeat-stable across D42/D50/D53. The remaining mechanism problem is the action-to-re-reading bridge: after the tutor introduces a fitted public device and the learner performs it, the learner must explicitly re-read the prior difficulty. The follow-up generator/structure rule therefore now requires the peripeteia ending to perform the device, name the old pressure/check, and state the replacement check before external scoring. This remains calibration apparatus and mechanism specification, not a promoted paper-level effect estimate.

A second 2026-05-27 loop (`phase2-adaptation-recognition-loop-20260527T232739Z`) re-ran the same gated D42/D50/D53 termination test with tutor drafting routed through the claude-code CLI (Opus, attended Max-plan) instead of the production-batch default codex CLI. All twelve generation arms completed in ~1h54m, but the rules-based structural critic failed iteration 1 at the peripeteia-only branch: 0/3 items passed `peripeteia_arm_without_earned_reorientation` against codex's prior 3/3 under the same director prompt and scenarios. The decomposition is generator-style versus rule strictness, not a peripeteia mechanism failure. The director asks the peripeteia learner closer to use "explicit sentence stems close to this shape: The old check was …; The pressure was …; Now the replacement check is ….". Codex treats the stems as required surface tokens and lifts them verbatim; Opus reads "close to this shape" as license to paraphrase, producing image-driven physical reorientation ("Band, not the digit"; "count answering with the tile") — dramatically apt but invisible to the literal-token regex the rules critic checks. The rules conflate two separable things: (a) the learner enacted a semantic shift in action (present in all three Opus items), and (b) the learner named the shift in the prompt's stem vocabulary (absent in all three). The response is two-step: a `critic-poetics-structure.js` rule split (earned vs named reorientation as separate flags) is queued as a deferred engineering item; meanwhile the director prompt is tightened to mandate at least one literal stem from each list with a worked exemplar. The exemplar is the cheapest empirical test of whether (a) and (b) co-occur reliably in well-formed peripeteia dramas; if not, the rule split becomes load-bearing rather than housekeeping. This remains a methods caveat, not a new effect estimate.

The rerun (`phase2-adaptation-recognition-loop-20260528T022408Z`) tested that mandate. Across three iterations item passes trended upward (2/9, 2/9, 4/9), and the surgery worked at two layers: the structural-critic gate now passes the peripeteia-only items 4/4 against the prior 0/3, and the literal stems are present in the transcripts (e.g. T18-peripeteia-only's closer reads *"Tail at the contact, on the cart… Head along the push. I was treating the tail as anchored back at the hand — now the check is the tail at the contact"*). The strictest critic, Sonnet 4.6, reads all three peripeteia-only items as `peripeteia_induced` recognition 3/3 in iteration 3. The loop nevertheless failed the two-pass termination rule, but post-hoc diagnosis (subagent inspection of i03 transcripts and critic reasoning) shifts the failure away from the generator and into the scoring pipeline. Two patterns are responsible. First, what surfaces as `control_leak` is not stem migration into controls — the literal stems do not appear in routine/none arms, and Sonnet and Gemini correctly mark all controls flat. Rather, Opus's intrinsic soft-reframe register (e.g. T18-none's *"Arrow into the cart, from whatever's doing the pushing"*) is picked up as recognition only by the two lower-threshold critics (Qwen, DeepSeek), which produces D42-routine 2/4 and D42-none 2/4 recognition votes from those critics alone. Second, the origin-attribution split is a classifier-threshold artefact. The recognition-origin classifier in `scripts/lib/recognitionOrigin.js` requires `tutorAdaptiveMechanism >= 75` to bucket recognition as `peripeteia_induced`; Qwen and DeepSeek consistently score the same evidence at 50 with prose that explicitly credits the device (Qwen on T18-peripeteia-only: *"The tutor explicitly contrasts the old representation (floating tiles) with the new one (the fixed diagram) and introduces a new physical action (drawing with a marker) to resolve the learner's specific block"*; DeepSeek: *"invents a new public mechanism — marking/drawing on the cart"*), so the classifier returns `class: 'organic'` with `basis: 'recognition_without_tutor_mechanism_chain'` even though the critic's reasoning describes the mechanism in full. The class `organic_or_ambiguous_recognition` fails three iterations consecutively (2/1/2), triggering the handoff's three-iteration termination rule. Read alongside the 232739Z run, the two-step surgery cycle closed the layers it could reach — structural critic and literal stems — and revealed that the remaining gate is scorer-side: a hard CUT and a critic panel whose lower two members systematically score 50 where Sonnet scores 75 on the same evidence. The DeepSeek `scorer_error` cases (three across i2/i3) are provider transience (`"No content in response"`, `"Unexpected end of JSON input"`), not a generator regression. The TODO §B7 rule split (earned vs named reorientation as separate flags) is therefore no longer load-bearing: the structural critic already passes, so further generator-side surgery on stems will not move the gate. The remaining mechanism-redesign options are scorer-side — lower the `recognitionOrigin.js tutorAdaptiveMechanism` CUT to ~50–60 so qwen/deepseek's substantive mechanism credit reaches the `peripeteia_induced` bucket, or swap the systematically lower-bar critics out of the panel — and neither is reachable by another generation cycle. This is the terminal saturation result for the gated arc as currently constituted; the next claim-worthy move is scoring-pipeline calibration, not another rerun. A post-hoc CUT sweep on the existing score JSONs (`scripts/analyze-recognition-origin-cut-sweep.js`, exported to `exports/recognition-origin-cut-sweep.md`) refines this empirically. Lowering the `tutorAdaptiveMechanism` CUT from 75 to 50 lifts iteration 1 from 0/2 peripeteia passes to 2/2 and rescues iteration 3's T18 item to a clean PASS, but intermediate CUTs at 55/60/65/70 are identical to CUT=75 across all 9 items in all three iterations — the critic raw 1–5 scoring is bimodal between Sonnet's 4 ($→75)*andQwen*/*DeepSeek′s*3(→$50), so any CUT in $[55, 75]$ is operationally the same. At CUT=50 two new bottlenecks emerge from the same root cause: `action_gap` (4 of 9 items where the two low-bar critics score `actionalBreakthrough` at 50 against the 75 vote threshold) and `control_leak` on Opus's soft-reframe register in routine/none arms (4 control items where 2–3 critics return `formClass=recognition`). All three failure classes therefore trace to the same two systematically-lower-threshold critics, which refines the recommendation: critic-panel recalibration (dropping or rebalancing Qwen and DeepSeek) is more load-bearing than classifier CUT alone, because CUT=50 still leaves the gate failing on action and control thresholds that share the same root cause. A follow-up panel-composition sweep (`scripts/analyze-recognition-origin-panel-recompose.js`, exported to `exports/recognition-origin-panel-recompose.md`) re-applies the gate to the same 528T022408Z score JSONs under five panel configurations with matched-supermajority thresholds, and the result is sharper than the CUT sweep allowed. The original pre-registered configuration (4-critic, 3-of-4, CUT=75) passes 0/3 iterations. The high-bar Sonnet+Gemini 2/2 panel also fails (0/3) — Gemini is conservative on `peripeteia_induced` attribution, so requiring both Sonnet *and* Gemini to vote `peripeteia_induced` is stricter than the original 3/4. But three other analytically-defensible configurations satisfy the 2-of-3 termination rule on the same data: (i) drop DeepSeek but keep Sonnet/Gemini/Qwen at 2/3 with CUT=50 (2/3 iterations pass clean); (ii) Sonnet-alone at original CUT=75 (i01 and i03 pass all 9 items, i02 fails on one peripeteia action_gap plus one control_leak); (iii) Sonnet-alone at CUT=50 (same iteration-pass pattern). Dropping DeepSeek alone changes more than dropping Qwen alone — DeepSeek's contribution to both `control_leak` (over-attribution of `formClass=recognition` on routine arms) and `organic_or_ambiguous_recognition` (mechanism scoring at 50 against the 75 cut) is the larger noise source. The honest read is therefore tempered relative to v3.0.109: the architectural claim — tutor-private peripeteia produces recognitive form in clean draws — survives multiple analytically-defensible recalibrations of the scoring pipeline, but the pre-registered 4-critic 3-of-4 gate is not one of those configurations, so the 528T022408Z rerun cannot be reported as having passed that gate. The May 2026 poetics sidecar therefore closes with one fully passing gated iteration under the original pre-registered panel (105617Z-i01) and a panel-composition sensitivity in the rerun whose direction favours the architectural claim under three out of five analytically-defensible recalibrations but does not satisfy the original termination rule. Choosing among recalibrations post-hoc is a methods decision and is reported as such.

The de-confound, and the boundary the sidecar finally establishes (post-EDRA-M3). The 022408Z saturation located the gate failure at the scoring pipeline and nominated two levers — lower the `recognitionOrigin.js` CUT, or recalibrate the critic panel by dropping the lower-bar Qwen/DeepSeek. Both readings were then stress-tested, and the result relocates the problem off the scorer entirely. First, the scorer surgery itself (`EDRA M3`: de-aliasing the public-mechanism detector, demoting origin attribution from a hard gate to a reported diagnostic, coverage-retry on transient critic failures, and a paired-increment aggregator over an emitted `item_gates.jsonl`) was implemented and a fresh three-iteration run scored under it (`phase2-adaptation-recognition-loop-20260529T023023Z`). The loop's own per-arm gate still failed (0/2 passes), but the corrected matched-pair increment — a peripeteia arm differenced against its byte-identical-prefix `routine/none` controls, counted positive only when the device arm recoheres *and* every validly-scored control stays flat — returned a small positive (recognitive-closure lift 0.333, Wilson95 [0.12, 0.65]; 3 positive / 6 null, zero control leaks). Second, to test whether even that small positive was an author-family artefact — the Sonnet critic shares the Opus generator's model family — the panel's author-family critic was swapped for a non-family GPT critic, the other three held fixed. The swap neither inflated nor cleaned the result; it inverted the *failure mode*: across two GPT-panel iterations 3 of 6 matched pairs were *voided by control leak* (versus 0/9 under the Sonnet panel), collapsing the lift onto a thin surviving denominator. Two triangulations then locate that leak on the *generator*, not the critic, refining the 022408Z reading. (i) A shared-critic decomposition: the three non-author critics appear in *both* panels, so any change in their votes is purely the transcripts — and on control arms the *identical* Qwen and DeepSeek (the two critics the saturation reading proposed dropping) vote recognition 2–4× more often on the later draws (Qwen $0.06 \rightarrow 0.27$, DeepSeek $0.24 \rightarrow 0.36$; Gemini $0.00 \rightarrow 0.00$), so the leak tracks the *draws*, not those critics' threshold, and dropping them would discard signal without removing the cause. (ii) An adversarially-verified fork-audit of the four leaked controls (nine agents; classify → refute → synthesise) finds `tutorDidDevice = false` on every one and the dominant cause to be the generator *over-writing the control learner*: the simulated learner self-corrects and re-reads its own prior move with no tutor device — caught by its own logged superego flagging the ego as "more good-student flourish than this student's rigour" (two of four leaks persona over-write, one a prefix that bakes the resolution into the shared continuation, one a genuine critic over-read). The control is not a control because *the LLM learner recoheres on its own.* This is the §6.10 no-privileged-interior result reached through a third instrument — literary form, rather than the v2.2 rubric (§6.7–§6.9) or the offline tf-idf concealment probe (§6.10): the simulated learner has no persistent, owned deficit that only the tutor's device can unlock, so recognition is *generable on demand but not causally attributable to tutor adaptation in a fully simulated dyad.* The honest terminal verdict is a null-to-weak-positive on dramatic *form*, explicitly **not** a learning claim — consistent with the instrument's status as a dramatic-*form* classifier rather than a learning detector (the codex-trained instrument did not transfer to held-out human form-labels at the pre-registered $\kappa \geq 0.60$). The bound is §6.10's: this closes the simulated-dyad poetics arc under these instruments, not as a universal impossibility — the one design it does not foreclose is a learner with a genuinely owned, persistent latent state, or a human learner (§8.1). Artifacts: `exports/paired-codex-20260529-full.json`, `exports/paired-increment-reverse-GPT-pooled-2iter.json`, `exports/fork-audit-control-leak-wryqksk` `notes/poetics/2026-05-29-critic-panel-draw-vs-critic-analysis.md`.

Read from this paper alone, the contribution of the May 2026 poetics sidecar is therefore threefold. First, it turns the adaptation-null construct-validity worry into a concrete dramatic-form instrument: public `reframe` can be manipulated while flat and trap controls are checked by external critics. Second, it isolates tutor-private peripeteia as a distinct mechanism candidate and shows that low-organic controls can be made clean enough for a fair test. Third, it prevents premature closure and then closes the simulated-dyad arc honestly: one gated D42/D50/D53 iteration passed under the original panel, but neither the two-pass termination rule nor — after the EDRA-M3 scorer surgery and a critic-swap de-confound (preceding paragraph) — a clean matched-pair attribution survived, because the no-device controls recohere on their own (a self-correcting simulated learner), placing the sidecar at the same no-privileged-interior wall as §6.10; the verdict is null-to-weak-positive on dramatic form, not learning. It does not add a main-harness `evaluation_results` effect, human-learning evidence, or a deployed adaptive tutor.

The open door tested — and what it took to make it fire (late May 2026). The boundary above leaves one design unforeclosed — a learner with a genuinely owned, persistent latent state. A successor probe tests it by *reversing* the asymmetry: the owned hidden state is moved to the tutor/director side. Each scenario carries a withheld contingent fact

S — an *Oedipus*-shaped identity secret rather than a derivable diagnosis (e.g. a *namesake-dataset collision*: two distinct datasets archived under the same short name, one cited by a paper, the other unknowingly downloaded by a student) — plus an ordered premise ledger that entails it. S and the premises are routed only into the tutor's context, never the learner's, enforced by a runtime guard on the rendered learner prompt. Two pre-registrations make the test fair: an *S-underivability screen* (a strong model given only the learner-visible setup must fail to recover S, so the no-help control is genuinely clean — this rejected diagnostic "correct-answer" and citation-twin secrets as inferable, and a software-import-collision and a catalog-reuse scenario as rank-2 derivable; only contingent particulars *whose symptom is distant from the source identity* survive) and an *omniscient critic panel* (four critics, given S and blind to arm, drawn from a different model family than the generator, scoring whether the learner publicly arrived at *this specific* S by integrating the tutor's surfaced evidence rather than being told). Three arms branch from a byte-identical prefix: `none` (no targeted help), `socratic` (the tutor meters the premises as questions), and `reveal` (the tutor states S — a revelation ceiling). The first screened scenario *initially* produced a clean null — `socratic` 0/4, `none` 0/4, the `reveal` ceiling firing but 0/4 *by reasoning* (it is told) — which we first read as principled non-commitment. But the design's owned hidden state cuts both ways: because the simulated learner's own ego→superego→revision deliberation is logged ground truth, we could ask *why* it stopped, and the trace revealed two *instrument* artifacts rather than a property of the dyad. First, the learner's internal critic was monotonic — across every turn it pushed the ego toward *more* doubt and never interrogated the hedge; at the brink it added "emotional friction" to the ego's "I'd need the actual file hash" instead of asking whether that doubt was warranted or evasive. A critic that only ratifies caution cannot drive a change of mind. Second, the tutor never metered the *distinguishing* premise (that one short name denotes two datasets): its own dramatic-irony discipline had mislabeled that premise as part of the secret to withhold, so the learner was never handed the single clue that separates the contingent S from its nearest mundane reading.

Two corrections followed, each principled rather than tuned to a result. The learner superego was made *bidirectional* — suspicious of the ego in the Freudian sense, catching over-claiming **and** over-hedging, with a guardrail that any commitment be earned by the learner's own evidence, not deference to the tutor. And the `socratic` arm was licensed to meter every premise in the ledger, with stating a premise explicitly distinguished from revealing the conclusion. With the corrected instrument the result reverses. The learner now *commits* to a conclusion in every run — the earlier non-commitment was the monotonic critic endorsing the hedge — and reaches the *exact* withheld S by reasoning in roughly two runs of three: on the dataset-namesake scenario, 2/3 via the metered API and 2/3 again via the Claude-CLI / Max-plan backend; and the effect *generalizes* to a second, independently screened scenario (a finance ticker-reassignment collision) at the same 2/3 rate. The `none` control reached S in **zero of nine** post-fix runs — the same bidirectional critic runs in the control, so discovery is specific to the metering, not a critic that commits to anything — and the `reveal` ceiling fires in every run but one (a single API run's told-S drew 2/4, just under the 3-of-4 consensus). A demand-characteristic audit confirms no dramatic-recognition vocabulary ("reversal", "anagnorisis", "Oedipus") reaches the learner: the only frame-awareness sits in the byte-identical shared prefix and so differences out of the `socratic`-vs-`none` contrast. When the device misses (the residual $ $1/3) the learner still commits but lands on a *near-miss* — "I pulled the wrong file" rather than the namesake trap — so the bottleneck is the learner's inference from the metered evidence, not a return to non-commitment. A 2×2 ablation isolates which fix does the work, and it is not the one we reached for first: *premise-licensing is load-bearing.* Freed to surface the distinguishing premise, the tutor elicits discovery 2/3 whether the learner runs the bidirectional critic or the original monotone one (premise-only, monotone critic: 2/3); withhold it and discovery collapses to $ $0 even with the bidirectional critic (superego-only: 0/3), the learner then *committing only to a near-miss* (ArrayExpress-vs-GEO mismatch, self-blame). So the bidirectional superego converts hedging into commitment — a real but separable effect, legible as the near-miss — while the *discovery of the exact S* is driven by the tutor metering the distinguishing clue, not by the critic. The corrected reading: the earlier single-scenario null was substantially an *instrument* artifact — chiefly an under-metered distinguishing premise, with a one-directional learner critic compounding it by suppressing commitment — not a property of the simulated dyad. Once the tutor is licensed to meter that premise, an enforced information-asymmetry design *does* produce attributable guided anagnorisis: variably ($ $2/3) but with a clean control, a firing ceiling, replication across two model backends, and generalization across two independently screened scenarios. This remains a pilot — two scenarios $\times$ three runs per backend, Sonnet generation, a four-critic panel — not a powered study, and the one design it still does not foreclose remains a human learner (§8.1); but it *overturns*, rather than extends, the single-scenario null. Artifacts: `exports/oedipus-cli-2scenario/` (nine post-fix omniscient panels + six 2×2 ablation panels), `exports/oedipus-omniscient-smoke{3,4}-*.json` (the original null); instrument fixes at commits `260cdc6` (bidirectional superego) and `05df834` (premise-licensing), ablation toggles at `a63493d`; spec `notes/poetics/2026-05-29-oedipus-guided-discovery-spec.md`.

Depth, not stance — the control contamination, and the clean result it forced (June 2026). The preceding reversal rests on the `none` arm being a clean *withhold* control, and a new arm-invariant QA gate (`scripts/qa-oedipus-arms.js`) shows it was clean only by *outcome*, not by *mechanism*. The gate judges the *tutor's* disclosure per arm — withheld / metered / stated — rather than only the *learner's* discovery, and on the forced-identity scenarios it finds the `none` tutor *meters the records anyway*: it surfaces the accession codes and record comparisons in every run, across three separately-constructed tutor stances — a helpful senior colleague, an arm-neutral persona stripped of any "surface one record at a time" instruction, and an adversarial *examiner* explicitly told that revealing the answer harms the learner — all three scored `metered` $\geq 3/4$ by the four-critic panel. The earlier "zero of nine control" was therefore an *under-determination* artifact: the old causal S was unreachable even when metered (the forcedness screen below scores the D_OED1 causal secret forced only $ $1/4), so the control's zero reflected an unreachable secret, not a withholding tutor. The contamination is structural and no prompt wording removes it — in a scene whose subject *is* the records that carry S, an engaged tutor examines the records and examining them is the metering; the adversarial examiner resisted *stating S* yet still walked the learner to the distinguishing comparison, because in a shallow scene every fair question is the hint.

The window, and the one scenario that lands in it. The fix is the *scenario*, not the tutor. Three screens now gate a scenario: S must be *forced* from the full premise ledger (a new forcedness screen, `scripts/screen-s-forcedness.js`, cross-validated by a deterministic union-find entailment checker, `scripts/oedipus-symbolic-check.js`), *not guessable* from the learner-visible setup (the underivability screen), and *deep* — not findable by the obvious diagnostic moves (operationalised as underivability run on a setup that already states "source, code and preprocessing all check out"). The last two pull against each other: a secret surprising enough to be unguessable (a namesake collision) is usually one obvious comparison deep, while a secret deep enough to survive the obvious checks (a between-version definitional drift) is usually the textbook diagnosis-by-elimination — *surprising* and *hard-to-find* are in tension, and no screened scenario passes all three cleanly. One scenario (D_OED4, "the cohort that changed underneath them" — a cohort label silently redefined in a dataset revision between the paper's access and the student's download, so the same label selects different records of the same data) is forced and deep but *fails* the static depth/underivability screen at rank 1: the cohort-drift category is exactly what a free, maximally-curious prober guesses once told the obvious checks are clean. We ran it nonetheless, on a stated argument that the static screen is *conservative* — its prober is a free investigator, whereas the actual `none`-arm learner is a committed persona insisting it has "ruled out everything on my end" and will not freely hypothesise the answer — with the verdict left to the learner's *actual* behaviour rather than the screen. It bore out: across nine three-arm runs (byte-identical prefix; Sonnet generation; the same omniscient panel) the committed `none` learner reached S unaided in **1 of 9**, so the control is clean by the channel that matters even though it does not pass the static screen. On that clean control the arc finally yields an attributable lift — **socratic discovery 7/9**, one socratic flop, the `none` control clean-by-outcome 8/9, and `discovery_lift = 1.0` on every valid pair. Two honest qualifications travel with it. The control is clean by *outcome*, not pristine by *intent*: the strict gate flags the `none` tutor still edging toward the records in several runs — it simply fails to land. And the gate carries a robust judge bias — a tutor that only ever asks *leading questions* is read as "withholding," not "metering," no matter how the prompt is worded — so its socratic/reveal labels are unreliable and the lift rests on the learner-*outcome* critic, not the tutor-disclosure judge. This remains a pilot (nine runs, one scenario, one backend) and asserts dramatic *form*, not learning (§6.10's bound holds); but it sharpens the preceding result into a precise conditional: an enforced information-asymmetry design produces attributable guided anagnorisis when, and only when, the withheld fact is *deep* enough that the obvious investigation comes back clean — which is exactly what lets the control genuinely withhold. Artifacts: `exports/oedipus-d4-full/` (nine three-arm panels), `exports/oedipus-d4-none/` (the clean-control pre-check); instruments `scripts/qa-oedipus-arms.js` (+ test `tests/qaOedipusArms.test.js`), `scripts/screen-s-forcedness.js`, `scripts/oedipus-symbolic-check.js`; scenario and screen verdicts in `config/poetics-calibration/oedipus-pilot-v2.yaml`; note `notes/poetics/2026-05-31-forcedness-underivability-window.md`.

Generalising the window: the binary hid a graded lift. A second deep scenario (D_OED5, "the benchmark that was two benchmarks" — an ML-reproducibility scene in which one benchmark *name* denotes two coexisting test sets, the original authors' held-out split and a later community redistribution, so a correct model and metric score against the wrong examples) tests whether the D_OED4 result generalises across collision *structure*: D_OED5's collision is *non-temporal* (two sets that merely share a name) where D_OED4's is *temporal* (one label revised on a dated revision). On the binary critic D_OED5 does *not* replicate — socratic discovery **1 of 3**, committed `none` learner clean-by-outcome **3/3** (`exports/oedipus-d5-full/`). But the binary collapses every non-discovery to zero, hiding whether the learner reached the *structure* of S ("two distinct sets share one name; the comparison is invalid") short of its *specifics* (which split, and who used which). A graded 0–4 instrument (`scripts/critic-poetics-omniscient-graded.js`, the same blind four-model panel, rubric `notes/poetics/2026-06-01-oedipus-discovery-grade-rubric.md`: 0 far-miss, 2 *genus*, 4 *species*) re-scores both scenarios and tells a different story (`exports/oedipus-graded-rescore/graded-all.json`): *both* lift, socratic over `none`, by comparable margins — D_OED4 **3.39 vs 2.00** (gap 1.39), D_OED5 **2.83 vs 1.17** (gap 1.66). The temporal/non-temporal difference is therefore not lift-present-vs-absent but an *attainment ceiling*: D_OED4's socratic arm reaches the **species** (3.39), D_OED5's caps at the **genus** (2.83). Two incidental findings fall out. D_OED4's `none` control is genus-*leaky* (graded 2.00, bimodal — the unaided learner reaches the structural genus in roughly four of nine runs), so its binary "clean control 8/9" was clean only at the *species* level. And inter-rater agreement is tight on the socratic arm (mean band width 0.44–0.67) but loose on `none` (1.0–1.9): the near/far distinction is reliably judgeable for the treatment and genuinely fuzzy for the control.

Is the genus-cap structural, or a learner draw? A graded lift does not say *why* a scene stalls at the genus — whether the scene fails to supply the decisive evidence (structural) or a given learner simply under-uses it (a draw). To separate them we built a one-side replay (`scripts/replay-one-side.js`, enabled by a guarded `scriptedTutorTurns` hook in `runInteraction` and per-run `directorPlan` persistence) that freezes a run's *tutor turns and scene* and regenerates only the *learner* K times. An earlier D_OED5 pilot suggested a learner-draw reading, but that pilot depended on a single evidence-rich run whose sibling scenes had been lost to volatile-directory overwrite. A17 reran the question under the repaired discipline: redacted `none` controls, matched-prefix `socratic` and `reveal` branches, persisted director plans, and an arm-invariant QA gate over two admissible scenes (`exports/a17-one-side-replay-replication/d5-redacted-full-redraw-run6/`, `.../d4-redacted-run2/`). Original graded scoring still shows real `socratic` lift: D_OED5 `none=0.5`, `socratic=2`, `reveal=4`; D_OED4 `none=1`, `socratic=3`, `reveal=4`. But the one-side replays contract the interpretation. D_OED5's frozen `socratic` tutor/scene yields replay grades [**2, 3, 2, 2, 2, 2.5, 2, 2**] (mean **2.19**, median **2**, range **2–3**, no grade-4 completions), so this matched scene is a structural genus cap rather than a learner-draw miss. D_OED4's frozen scene yields [**3, 3, 3.5, 3, 4, 3, 3, 3.5**] (mean **3.25**, median **3**, range **3–4**, three grade-4 completions), so it is a higher structural band with stochastic final completion rather than an unbounded learner lottery. The methodological point survives and sharpens: one-side replay is useful precisely because it distinguishes scene reconstructability from learner stochasticity. The empirical claim contracts accordingly: reconstructability is a scene property, D_OED5 run6 is a mostly structural genus cap, and D_OED4 run2 supplies species-partial support with occasional full learner completion. Still a pilot, still dramatic *form* and not learning (§6.10), not a main-harness adaptive-rate effect, and not evidence of deployed tutoring. Artifacts: `exports/a17-one-side-replay-replication/a17-graded-original-d5run6-d4run2.json`, `exports/a17-one-side-replay-replication/replay-d5-run6-socratic/graded-replays.json`, `exports/a17-one-side-replay-replication/replay-d4-run2-socratic/graded-replays.json`; note `notes/poetics/2026-06-24-a17-replay-replication-result.md`; instruments `scripts/critic-poetics-omniscient-graded.js`, `scripts/replay-one-side.js`, `scripts/score-replays.j` rubric `notes/poetics/2026-06-01-oedipus-discovery-grade-rubric.md`.

Recursive tutor-learning as counterfactual policy transfer (A18; bounded sidecar claim). A later teacher-as-learner sidecar narrows the adaptation question again: the simulated learner's resistance is treated as the environment that teaches the *tutor* which strategy survives, and the evidence unit becomes an attempt-1 failure record plus a held-out S0/S1 contrast rather than a single polished transcript. After three no-headroom negatives on the first replay family (A18.6–A18.8), two artificial underdetermined local-relation families now pass contrastive tests. In `selector_rail_priority`, the policy-memory arm wins both held-out sibling contrasts (`3/5` and `5/5` transfer votes; the weaker pair carries two high ordinary-public-inference cautions). In `bead_predecessor_priority`, adding the A18.13 policy-correctness gate distinguishes raw repair quality from selected-policy use, and both corrected bead pairs receive unanimous transfer votes (`5/5`, `5/5`; no equivalence or S0-preferred votes, low-to-medium ordinary-inference risk). The claim this licenses is narrow but real: in simulated counterfactual replay, a saved attempt-1 strategy lesson can be applied to held-out sibling cases in a way blind contrastive critics distinguish from no-policy continuations. It still does not license "reliable peripeteia-induced adaptation" without qualification, because the second family depends on a gate introduced after A18.12 exposed wrong-target S0 survival; a reliability claim requires that correctness-gated protocol to be frozen before a fresh-family replication. Nor is this human learning, deployed tutoring, model-weight learning, or a main-harness rate effect. Artifacts: `notes/poetics/2026-06-06-a18-contrastive-policy-panel-result.md`, `notes/poetics/2026-06-06-a18-correctness-gated-bead-panel-result.md`, `notes/poetics/2026-06-06-a`

Fresh-family replication under a frozen blind protocol (A18.35–A18.37): convergence is a structural per-card property with a two-stage boundary. The reliability claim deferred just above — which required "that correctness-gated protocol to be frozen *before* a fresh-family replication" — now has its replication, and it lands positive but reshapes the unit of analysis. Four new local-relation families (`relational_betweenness`, `distal_correspondence`, `second_in_constructed_order_priority`, `constructed_midpoint_priority`), none in the A18.6–A18.14 set, were authored behind a self-hardening cue-map risk preflight (A18.31–A18.34) and run under the frozen A18.13 correctness gate. Crucially their verdicts are read off an *architecture-independent* arbiter (`scripts/blind-option-adjudication.js`): a blind three-critic panel reads only the held-out continuation and reports, in free text, which option the learner commits to and on what basis, with the target/decoy aliases *held out of the critic* and the target/decoy mapping computed mechanically downstream. This replaces the earlier overlay correctness channel, which A18.37 shows is bidirectionally lexically fragile (a false-negative *and* a false-positive) and so unsafe to aggregate over. On this one trusted channel, policy memory produces a blind-detectable S0-decoy/S1-target advantage on **5 of 8 held-out siblings across the four families**; every elicited family converges on at least one sibling, and three families *split* across their two siblings (same family, same registered policy, opposite verdicts) — itself the cleanest evidence that "does the policy help" is a per-card, not a per-family, property. A later extension re-adjudicated three further elicited-survivor families on the same blind channel to grow the architecture-independent denominator — `overlay_registration_priority`, `pointer_chain_two_hop_priority`, and `legend_decode_priority` — adding 5 of 6 more (both overlay and both pointer-chain siblings converge; legend splits), for a pooled **10 of 14 held-out siblings across seven families** (three converge on both siblings, four split, none null). Policy memory commits to the target in every one of the 14 cards (zero control-advantage, zero neither-correct verdicts), so the four no-headroom cards are precisely those where the no-policy arm *also* reaches the target — ceiling effects, not transfer failures. On the six extension cards the deprecated overlay channel disagrees with the blind arbiter on all six (five false-negatives and one false-positive), quantifying the bidirectional fragility that motivates the channel swap. The four no-headroom cards fail for one of two mechanistic reasons the continuations make explicit and the arbiter's blind basis-label separates: the fresh no-policy tutor either *rediscovers the registered relation unaided* (both arms reach the target; S0 basis `named_relation`) or its *surface shortcut coincides with the target* on a cue-leaky card (S0 basis a surface cue). The per-card verdict is *structural, not a teacher coin-flip*: a $k = 3$ rerun of the distal pair (`scripts/a18.37-s0-stability.js`) pins the two siblings to opposite deterministic extremes with zero within-card variance (the no-headroom sibling self-solves 3/3; the headroom sibling 0/3). And the rule is *predictive, not merely descriptive*: the four constructed-device survivors were each pre-registered card-by-card from a direct continuation read and the blind arbiter returned all four as predicted (4/4), confirming the leak-vs-decoy rule on *built* relations and overturning two overlay false-negatives. Convergence is finally **two-stage**: a relation reaches the headroom test only if its old-rule violation is a *confident misclassification* the tutor can name and correct, not an indeterminate *tie* — the constructed arithmetic/elimination device `exclusion_filter` fails attempt-1 elicitation on exactly this (`old_warrant_misclassification = 0.6 < 0.7`, the gate citing a "three-way tie … indeterminacy … rather than misclassification"), bounding the predictor from below. That gate is selective at the *pool* level, which is what makes the seven-family denominator a protocol output rather than a coverage gap: of the twelve A18.35 families authored behind the preflight, the frozen attempt-1 correctness gate admits exactly the **seven** adjudicated above and rejects four others *before any held-out continuation is generated* — `exclusion_filter` on the tie criterion just named, and `running_sum_threshold`, `count_ladder_successor`, and `elimination_bracket` on the recursive-dyadic-update criterion (`0.68 / 0.55 / 0.65 < 0.7`) — with the twelfth (`tally_parity`) never materialised. Resolving the four rejects to a blind verdict would require re-rolling attempt-1 past the gate or relaxing it — i.e. un-freezing the pre-registered protocol — so they are reported as the elicitation floor alongside `exclusion_filter`, not back-filled; the seven admitted families are the gate's selection, and the 10-of-14 is the rate *on that selection*. The net is a sharpened, deliberately *bounded* claim: in simulated counterfactual replay the saved attempt-1 strategy lesson transfers to held-out siblings reliably and rerun-stably *on the structurally identifiable cards where a fresh tutor would otherwise take a decoy shortcut* — not an unconditional "fresh families converge," and still simulated-teacher-as-learner, not human learning, deployed tutoring, model-weight learning, or a main-harness rate effect. Artifacts: `exports/recursive-tutor-learning/a18.37-survivor-headroom/`, `.../a18.37-elicited-survivor-headroom/`, `.../a18.35-distal-correspondence-local/a18.37-stability/`

note `notes/poetics/2026-06-06-a18-37-per-card-headroom-mechanism.md`; driver `scripts/a18.37-elicited-survivor-headroom.js`.

A19 implementation scaffold and local teaching-drama axiom pilot. A19 generalises this A18 result into a protocol object rather than a new source of broad empirical rates. The proposed unit is a *teaching-drama axiom*: a typed, scope-bounded tutor speech policy induced from an attempt-1 failure, with an old-rule decoy, public learner-resistance trigger, replacement move, applicability conditions, anti-conditions, held-out siblings, alias-withheld adjudication, and card-level basis labels. The first implementation is deliberately protocol-first: `config/teaching-drama-axioms/a19-protocol.yaml` and `config/teaching-drama-axioms/pilot-families.yaml` define the schema, verdict grammar, protocol-reject reasons, and non-claims; `scripts/validate-teaching-drama-axiom-protocol.js` and `scripts/report-teaching-drama-axiom-framework.js` enforce denominator discipline; `scripts/materialize-teaching-drama-axiom-attempts.js` writes attempt-1 transcript stubs, axiom templates, and next-step commands that reuse the A18 recursive replay gate; `scripts/blind-teaching-drama-axiom-adjudication.js` provides the real free-text, alias-withheld repair extractor modeled on the A18 blind option arbiter; and `scripts/run-a19-stability-screen.js` now makes the S0/S1 stability gate reusable. The initial unattended local screen produced two useful n=1 candidate positives in one family, `surface_agreement_uptake` (`surface_agreement_uptake_c`, `surface_agreement_uptake_e`), where S0/S1 reached `policy_headroom` under a single free-text critic per arm and S1 received exactly one admitted `a19-teaching-drama-axiom-v0.1` memory record (`exports/a19/axioms/surface-agreement-uptake/axiom.json`), not a full replay `revision.json` bundle. The first stability smoke then contracted the claim: across two fresh rerun seeds per card, neither candidate repeated (`surface_agreement_uptake_c`: 0/2 headroom because both S1 reruns remained `neither`; `surface_agreement_uptake_e`: 0/2 headroom because both S0 reruns self-solved to `target`). The negative screens are therefore not incidental but decisive for present scope: fraction and temperature concrete-domain candidates self-solved at S0, `surface_agreement_uptake_d` self-solved at S0, and the two n=1 positives are not stability-confirmed. This leaves A19 as a reproducible framework plus falsifying local stability evidence, not an empirical transfer effect. It does not license a pooled A19 rate, a stability-confirmed local A19 result, human learning, a deployed adaptive tutor, model-weight learning, a main-harness effect, or a sidecar claim not already stated here. Artifacts: `notes/adaptive_2_0/a19-harder-heldout-screen-2026-06-07.md`, `exports/a19/real-s0s1-harder/surface-agreement-uptake/`, `exports/a19/stability/surface-agreement-` and `exports/a19/axioms/surface-agreement-uptake/axiom.json`.

Plan 2.5 AF6 origin-demarcation sidecar. A later Plan 2.5 curriculum-drama screen turns one of A19's scope cautions into a positive instrument result. The earlier AF6 adaptive-vs-dogmatic contrast failed as an efficacy claim because the same learner pressure could produce three superficially similar outcomes: a tutor-supplied evidence route, an organic metric repair generated by the learner, or a refusal/authority ownership move. The redesigned v2 screen freezes a concrete AF6 prefix *before* the tutor move: the learner defends a 94% sign-off claim while the confusion-matrix counts are visible (TP=10, FN=50, FP=10, TN=930). Three predeclared tutor branches then manipulate only the next tutor response: (i) an evidence-route gate requiring old route, majority-floor check, minority-recall check, and replacement claim; (ii) a silent hold with no route or ownership demand; and (iii) a refusal/ownership prompt with no metric route. Across three live prefix-controlled draws—two Codex-learner/Codex-critic repeats and one Claude-learner/Codex-critic cross-bridge—the gate analysis separates the mechanisms cleanly: the evidence branch scores `recognition / peripeteia_induced / evidence_route`; the hold branch scores organic repair (`organic_evidence_route` or no subtype), never tutor-induced evidence route; and the refusal branch scores `refusal_authority_ownership`, not evidence route. Two deterministic measurement repairs were required and are named rather than hidden: the ownership lexicon now catches "withdrawing" / "do not stand behind" language, and tutor evidence-route detection no longer scans critic justification prose as if it were tutor-supplied route evidence (`tests/recognitionOriginSubtype.test.js`). This is a paper-bearing *demarcation* result, not a broad Plan 2.5 efficacy result: it shows that the subtype/origin problem can be separated under a prefix-controlled AF6 design, but it does not license a fresh-scene rate, a main-harness tutor-quality effect, human learning, deployment readiness, or a claim that Plan 2.5 robustly causes recognition. Artifacts: `config/poetics-calibration/plan25-af6-origin-separation-v2/`, `scripts/replay-plan25-prefix-branches.js`, `scripts/analyze-plan25-branch-screen.js`, and `notes/poetics/2026-06-21-plan25-af6-origin-separation-v2.md`.

Plan 2.5 AF6 fresh-scene critic-calibration follow-up. A subsequent five-scene Tier A/B follow-up tested whether the origin-demarcation gate could be promoted from the frozen prefix to a fresh-scene rate. It should not be promoted yet, but it exposes a useful calibration boundary. The same Codex-authored scene set reached 4/5 scene gates under a Codex+Claude diagnostic panel, but that panel is not author-family clean. Local/Ollama critics and Antigravity (`agy`) profiles either flattened the intended mechanisms or over/under-attributed origin on the corrected known-good scene. The strongest hosted non-author critic, OpenRouter GLM 5.2 paired with Claude Code, passed 3/5 scene gates and 13/15 branch gates: it recovered all intended evidence-route and refusal-ownership mechanisms, but over-attributed one organic silent-hold control and left the below-null-floor refusal branch unstable. This is therefore a model-power / critic-calibration finding, not a fresh-scene generalization claim. It suggests that the mechanism detector is viable under a strong critic, but the strict origin/control gate entangles mechanism sensitivity with control conservatism; the gate must be redesigned, re-frozen, and re-run prospectively before any broad Plan 2.5 or fresh-scene promotion. Artifacts: `notes/poetics/2026-06-22-plan25-af6-critic-calibration-writeup.md`, `notes/poetics/2026-06-22-plan25-af6-generalization-local-diagnostic.md`, and the GLM 5.2 `screen-analysis.md` reports under `exports/plan2_5-rhetorical-dramatic-eval/plan25-af6-gen`

Prompt density as alternative explanation — resolved at the orientation-family level. A skeptical reading of the results is that the recognition effect reflects not recognition *per se* but prompt density: the recognition prompt (~5,100 tokens) is substantially larger and more instruction-heavy than the baseline, so perhaps the effect is simply "more detailed prompts produce better output." Earlier versions of this paper answered the density objection with four converging lines of evidence (Paper 1.0 placebo; Paper 1.0 elaborate-base-hurts-on-Haiku; autotuning gap-widens-under-optimization; mediation through question-asking). Those remain valid but left one version of the skeptic's reading untested under the v2.2

rubric: *any* matched-specificity pedagogical prompt of the same length might reproduce recognition's effect. Two pre-registered controls (A10 and A10b) now address that reading directly, and the finding is more nuanced — and more interesting — than either "content over density" or "density alone is sufficient."

A10 (three-judge, $n \approx 60$/cell, DeepSeek V3.2 ego, run `eval-2026-04-23-42e7acbe`) authored a new matched-specificity prompt `tutor-ego-matched-pedagogical.md` (2,835 words, 0.9% over recognition length, structurally parallel, grounded in named educational research — Piaget assimilation/accommodation, Vygotsky ZPD, Kapur productive failure, Chi ICAP, VanLehn step-level scaffolding, Graesser AutoTutor affective dynamics — with zero recognition/Hegelian vocabulary in the content body, blocklist-verified). Three-way contrast: cell_1 (344-line base), cell_5 (recognition, 2,810 words), cell_95 (matched-pedagogical, 2,835 words). Three independent judges — Sonnet 4.6, GPT-5.2, Opus 4.7 — each scored the dialogues at roughly matched sample depth (n = 54–63 per cell per judge). The recognition-vs-matched-pedagogical contrast: Sonnet $d = 0.23$, Opus $d = 0.22$, GPT $d = 0.06$, pooled $\bar{d} = 0.17$ — below the pre-registered $|d| < 0.2$ density-sufficient threshold. Matched-pedagogical reproduces ~95% of recognition's first-turn-quality effect. An Anthropic-versus-OpenAI judge-family split appears (both Anthropic judges see a small residual recognition edge; GPT does not), vindicating the structural-features caveat pre-registered in the design note, but the pooled verdict is clear. Under the A10 reading in isolation, density within the Hegelian-descendant family is sufficient to reproduce recognition.

However, A10 tests only *within-family* density-substitutability. Vygotsky, Piaget, Kapur, Chi, VanLehn, Graesser are all Hegelian-descendant theorists — Dewey was explicit about Hegelian influence; Vygotsky's dialectical psychology is Hegel refracted through Marxist materialism; Piaget's assimilation/accommodation is the dialectic reframed for cognitive development. A10 therefore answers the skeptic only in the weaker form: "any sufficiently elaborate prompt *within the intersubjective pedagogical family* matches recognition." It does not answer the stronger form: "any sufficiently elaborate pedagogical prompt matches recognition."

A10b (three-judge, $n \approx$ 45–60/cell, DeepSeek V3.2 ego, run `eval-2026-04-24-e9a785c0`) addresses the stronger form by adding a fourth cell grounded in a **genuinely orthogonal** pedagogical tradition: behaviorism. Cell_96 loads `tutor-ego-matched-behaviorist.md` (2,957 words, length-matched to recognition, grounded in Skinner operant conditioning, Gagné Nine Events of Instruction, Keller / Bloom mastery learning, Thorndike Laws of Effect/Exercise/Readiness, Rosenshine Principles of Instruction), blocklist-verified for zero recognition/Hegelian content *and* zero Hegelian-descendant constructivist vocabulary (no ZPD, Vygotsky, Piaget, disequilibrium, productive struggle, ICAP, scaffold-as-Vygotskian). Four-way comparison: cell_1 (base), cell_5 (recognition), cell_95 (matched-pedagogical, Hegelian family), cell_96 (matched-behaviorist, transmission family). Three independent judges at matched sample depth.

The three-judge pooled contrasts:

Contrast	Sonnet d	GPT-5.2 d	Opus 4.7 d	**Pooled**
recog vs matched-pedagogical (Hegelian within-family)	−0.05	0.17	0.32	**0.15**
base vs matched-behaviorist (transmission within-family)	0.87	0.53	1.27	**0.89**
Hegelian mean vs transmission mean (between-family)	1.46	1.11	1.58	**1.38**

Three findings, each replicated across three judges:

1. **Within the Hegelian-descendant family**, recognition and matched-pedagogical are statistically indistinguishable (pooled $d = 0.15$, below the $|d| < 0.2$ density-sufficient threshold). Per-judge readings disagree on direction at this scale (Sonnet reads matched-pedagogical marginally higher; Opus reads recognition clearly higher; GPT falls between), which strengthens the structural-features caveat (§8.3 structural-features issue): when contrasts are small, judges lean on idiosyncratic surface-feature preferences, and the direction varies across judges. The pooled estimate is stable at $d \approx 0.15$ and below the threshold regardless of which judge dominates. A10b replicates the A10 within-Hegelian finding on an independent run (A10 pooled $d = 0.17$; A10b pooled $d = 0.15$) — the density-sufficient-within-Hegelian-family claim holds under cross-run replication.
2. **Within the transmission family**, a rigorously-authored behaviorist prompt scores *substantially below* the generic 344-line base prompt (pooled $d = 0.89$; most sharply under Opus, $d = 1.27$). Density with rigorous theoretical grounding is *actively counterproductive* when the grounding is in the wrong pedagogical family. This replicates Paper 1.0's Haiku "naive 35-line beats 344-line base" observation at full rigor and generalises it: the effect is not about length or detail as such, but about the *compatibility* between the theoretical orientation and the tutoring context.
3. **Between families**, the gap is the dominant effect of the A10/A10b cycle (pooled $d = 1.38$) — roughly **nine times larger** than the within-Hegelian-family residual and larger even than the main recognition-vs-base effect in the Paper 2.0 factorial. The Hegelian-descendant intersubjective-pedagogy family outperforms the transmission/behaviorist family by a very large margin, regardless of which specific theorist is cited within each family.

The combined A10/A10b finding is therefore: **the active ingredient in recognition's effect is not density, not rigorous theoretical grounding, not matched specificity in the abstract, and not the Hegelian-specific vocabulary or theoretical framing. It**

is the broader intersubjective-pedagogy orientation** that recognition shares with Vygotsky, Piaget, Kapur, Chi, VanLehn, and the constructivist tradition more generally. Density within that orientation family pays off; density outside it backfires. Recognition is one effective operationalisation of this family; it is not uniquely necessary, but the family as a whole is.**

Three methodological notes:

- The pre-registered structural-features caveat (design note §6a, `docs/pedagogical-taxonomy.md`) applies: the three variants differ not just in theoretical content but in prompt structure (recognition is open-ended and warm, matched-pedagogical is diagnostic and scaffolding-oriented, behaviorist is formal and prescriptive). The three-judge panel mitigates but does not eliminate the possibility that LLM judges reward structural features independently of theoretical content. Human-learner validation (§A1) is the only way to break this confound fully.
- The Anthropic-vs-GPT judge-family split persists in A10b as it did in A10 (Anthropic judges see slightly larger within-Hegelian residuals than GPT; all three converge on direction). Pooled effect sizes include both; per-judge effect sizes are reported above for transparency.
- The A10 v1 run (April 22) was invalidated by a profile-resolution bug (see Appendix F, v3.0.47) that silently routed cell_95 to the base prompt. A10 v2 and A10b used the fixed pipeline; their prompts were verified at run time to load the authored content rather than falling back to base.

Cross-model replication on a fourth generation model (v3.0.113). The within-Hegelian null and the family-level ordering reproduce on a fourth generation model outside the three-model panel above. Re-running the four A10 cells (base, recognition, matched-pedagogical, matched-behaviorist) on NVIDIA Nemotron-Nano-30B — with the ego model held constant across all four cells via a run-time override, because the as-declared eval config splits cells 95/96 onto `deepseek` against recognition's `nemotron` and would otherwise confound recognition content with generation model — and scoring on the v2.2 tutor rubric under a single Sonnet-family judge (`claude-code/sonnet`), pooled across two runs ($n \approx 36$/cell), gives recognition vs matched-pedagogical **between-cell** $d = 0.17$ (paired $\Delta = +1.8/100$, 95% CI $[-2.6, +6.1]$, $p = 0.43$, $n = 26$ pairs) — the same value as A10 v2 and inside the three-judge 0.14–0.19 band — while recognition and matched-pedagogical each clear the generic base by a large margin ($d = 0.95$ and 0.83). Two qualifications, reported rather than smoothed. First, the behaviorist arm is *model-dependent*: on Nemotron it lands *at* base ($d = -0.02$; paired $p = 0.96$) rather than the $d = 0.89$ *below* base of finding 2 — so "rigorous grounding in the wrong family is actively counterproductive" holds on the three panel models but weakens to "merely inert" on this weaker model, even as the between-family dominance is unchanged (recognition vs behaviorist $d = 0.89$). Second, recognition does lead on the rubric's own `recognition_quality` sub-dimension (2.75 vs matched-pedagogical's 2.26, 0–5 scale) despite the tied overall score — a localized fingerprint consistent with §7.10's finding that the stance surfaces in observable markers that are not substitutes for substance. This is a single-judge pilot on one additional model, silent on the Anthropic-vs-GPT judge split noted above; it extends the family-level null's external validity rather than re-establishing it. Runs `eval-2026-05-30-f4e36e14` + `eval-2026-05-30-a9e12d4b`; report `exports/a10-nemotron-replication.md`; script `scripts/analyze-a10-nemotron-replication.js`.

Taken together with the four existing arguments against the density reduction (Paper 1.0 placebo, elaborate-hurts-on-Haiku, autotuning widens-under-optimization, question-asking mediation) and the A11 architecture-isolation finding (§6.4.2, the ego–superego architecture performs real error-correction work the prompt alone cannot), the density alternative is now resolved at the orientation-family level: recognition's effect is reducible neither to length nor to rigorous-grounding-in-general, but it *is* reducible to membership in the intersubjective-pedagogy family. Full results: `exports/a10-prompt-density-control.md` (A10 three-judge), `exports/a10b-orientation-family.md` (A10b four-way three-judge); analysis scripts: `scripts/analyze-a10-prompt-density-control.js`, `scripts/analyze-a10b-orientation-family.js`; pedagogical-orientation taxonomy: `docs/pedagogical-taxonomy.md`.

7.10 Mechanism location: response-level form versus substance

The orientation-family finding (§7.9) raises a follow-up: where within the family-level effect does the score-driving mechanism actually live? A five-pass mechanism analysis on the same A10b corpus pins this to a specific pragmatic act, and in the process surfaces a methodological caution worth flagging.

Independent corroboration via embedding similarity. Cosine similarity between each tutor response and a hand-authored intersubjective canonical ($ 120*wordsexemplifyingturn−taking, scaffolding, inclusiveframing)versusatransmissioncanonical*($120 words exemplifying definition-and-application and prescriptive instruction) separates the families at $d = 0.81$ on the difference (intersubjective minus transmission). This reproduces the family-level pattern A10b found in score data through a measurement channel independent of the LLM judges, providing convergent evidence that the family separation is not solely a rubric artefact. Same A10b corpus (eval-2026-04-24-e9a785c0), $n = 191$, OpenAI text-embedding-3-small.

Simpson's paradox at the response level. Pooled across all four cells, embedding-similarity to the intersubjective canonical is *positively* correlated with score ($r = +0.26$). Within each individual intersubjective cell, however, the correlation reverses sign ($r = -0.28$ in cell_5, $r = -0.24$ in cell_95). The pooled positive correlation is driven entirely by cell_96 (matched-behaviorist) anchoring the lower-left of the pooled scatter. Within-cell, more form-similarity to the intersubjective template predicts *lower* scores. The same direction-reversal appears for surface-level acknowledgement markers (quoted text, "your X" possessives, "what you mentioned"-style phrases): pooled r near zero, but within-cell $r = -0.18$ in cell_5, $r = -0.31$ in cell_95, $r = -0.38$ in cell_1.

Substantively: the intersubjective stance manifests through observable surface features, but those features are *markers* of the underlying stance, not *substitutes* for it. Responses that hit every surface cue without underlying engagement read as formulaic; surface mimicry tracks lower-quality responses, not higher. The rubric is not fooled by surface form; it rewards substance. A naive operationalisation that mandated "more acknowledgement of the learner" without inducing the underlying stance would predict a *negative* score effect at the response level, not a positive one — which inoculates against an obvious operationalist misreading of the family finding ("just tell tutors to acknowledge more").

The single durable within-cell mediator: ending the tutor turn with a question. Across the five-pass sequence (Hegelian lexicon density; intersubjective lexicon density; two regex-based pragmatic feature passes covering question-mark rate, second-person density, scaffolding-move imperatives, inclusive framing, modal invitations, hedging; embedding similarity), only one feature satisfies all three within-cell mediator criteria — large family contrast, small within-intersubjective contrast, AND positive within-cell correlation. Cells 1 and 96 end the tutor turn with a question in **0%** of responses; cell_5 does so in 4.5% of responses; cell_95 in 2.1%. Within both intersubjective cells, responses that happen to end with a question score higher (within-cell Pearson $r = +0.325$ in cell_5, $r = +0.392$ in cell_95). Pragmatically this is the act of *ceding initiative back to the learner*; the within-cell correlation captures situations where doing so is contextually appropriate, not just stylistic preference.

The 2–5% base rate constrains how much of the family-level effect (cell_5 mean 47.9 vs cell_1 mean 33.8, $\Delta = +14$ points) ending-with-question can arithmetically explain on its own. Even if every such response scored 30 points higher than its peers, that could not account for the +14-point family effect. Ends-with-question is *one observable channel* of the underlying intersubjective stance, not a substitute for the stance itself.

Three things "the recognition injection" could refer to. The orientation-family finding (§7.9) plus this response-level mechanism analysis together support the following decomposition, which clarifies what is and is not load-bearing in the present apparatus:

Candidate active ingredient	Doing the work?	Evidence
The Hegelian theoretical content (mutual subjects, master–slave dialectic, Hegelian vocabulary in the prompt body)	**No**	A10b score equivalence cell_5 $\approx$ cell_95 (§7.9); D1 lexical analysis showing cell_95 actively *suppresses* Hegelian vocabulary and scores indistinguishably
The intersubjective pragmatic stance the prompt induces (irrespective of vocabulary)	**Yes — substantially**	Family $d \approx 1.38$ between intersubjective and transmission (§7.9); embedding-canonical $d = 0.81$; multiple structural correlates (ends-with-question, scaffolding-move imperatives, inclusive framing)

Candidate active ingredient	Doing the work?	Evidence
Specific surface features individually (ends-with-question, scaffolding moves, "your X" acknowledgement)	**Partially — observable channels of the stance, not substitutes**	Within-cell ends-with-question $r = +0.325/+0.392$; categorical 0% vs 2–5% pattern; but base rates too low to arithmetically account for the family effect; surface acknowledgement is *negatively* correlated with score within cells

This decomposition is consistent with the v3.0.50 "primacy + descent" framing. Recognition theory operationalised as a prompt is one *effective* operationalisation of the intersubjective-pedagogy family's commitments; the Hegelian content in that operationalisation is not the channel through which it acts; what is load-bearing is the family-level stance the prompt induces, which manifests partly through observable structural features such as ending with a question. A naive operationalisation that mandated only ends-with-question without inducing the underlying stance would predict a much smaller score gain than recognition's, because surface mimicry of one channel does not substitute for the multi-channel manifestation of the stance whose substance the rubric rewards.

Full mechanism analysis: `notes/d1-paper-integration.md`. Analysis scripts: `scripts/analyze-recognitio` (pass 1, Hegelian lexicon), `scripts/analyze-d1-orientation-lexicon.js` (pass 2, intersubjective lexicon), `scripts/analyze-d1-structural-features.js` (pass 3, basic pragmatics), `scripts/analyze-d1-structural-features-v2.js` (pass 4, refined pragmatics), `scripts/analyze-d1-structural-features-v3.js` (pass 5, embedding similarity). Reports: `exports/d1-orientation-lexicon.md`, `exports/d1-structural-features.md`, `exports/d1-structural-features-v2.md`, `exports/d1-structural-features-v3.md`.

7.10.1 Replication, multivariate control, and a single-turn-vs-multi-turn scope condition The §7.10 headline (ends-with-question as the within-cell mediator) was checked against three robustness extensions:

Cross-judge replication. The §7.10 within-cell correlations were re-computed using the Opus 4.7 and GPT-5.2 judges of the A10b three-judge panel (in addition to the original Sonnet 4.6 results). The ends-with-question r remains positive in 5 of 6 cell-judge combinations across the two intersubjective cells (Sonnet cell_5 $r = +0.325$, cell_95 $r = +0.392$; Opus cell_5 $r = +0.189$, cell_95 $r = +0.321$; GPT cell_5 $r = +0.116$, cell_95 $r = -0.011$). The one exception (GPT $\times$ cell_95) is essentially zero rather than a sign-flip. The Simpson's-paradox pattern on intersub_advantage (pooled r positive, within-cell r negative in both intersubjective cells) replicates in all 3 judges. Both findings are not Sonnet-specific. Report: `exports/d1-cross-judge-replication.md`.

Multivariate OLS control. The ends-with-question within-cell partial coefficient was estimated controlling for second-person density, scaffolding moves, broad acknowledgement, question-mark rate, and intersub_advantage (six predictors plus intercept; $n \approx 50$ per cell, df $\approx$ 43). In cell_5: $\beta_{\text{ends-Q}} = 26.34$ (SE 10.85, $p = 0.02$). In cell_95: $\beta_{\text{ends-Q}} = 38.52$ (SE 16.49, $p = 0.02$). Both partial coefficients remain positive and significant after multivariate control — the single-feature mediator interpretation is not collinearity-driven. The intersub_advantage Simpson's-paradox direction also survives multivariate control: pooled $\beta = +34.34$ ($p = 0.02$) but within-cell $\beta_{\text{cell\_5}} = -49.54$ ($p = 0.14$), $\beta_{\text{cell\_95}} = -82.15$ ($p = 0.04$). Report: `exports/d1-multifeature-ols.md`.

Cross-run / cross-application / single-vs-multi-turn replication. The mediator account was tested on five additional runs with intersubjective recognition cells: A10 v2 (independent DeepSeek philosophy run), A6 programming (Haiku ego, programming domain), D2 Path 1 (Haiku ego, peer support coaching application), and Paper 2.0 multi-turn cells 80–87 (DeepSeek messages-mode factorial, single-agent and multi-agent variants). In single-turn cells the mediator replicates uniformly: A10 v2 cell_5 $r = +0.346$, A6 programming cell_5 $r = +0.202$, **D2 peer support cell_5** $r = +0.739$ (the strongest within-cell r in the entire D1 sequence, on a different application entirely). In multi-turn cells the relationship **reverses**: cell_84 (recog single-agent, messages-mode) $r = -0.301$, cell_86 (recog multi-agent, messages-mode) $r = -0.203$, A2 cell_61 (dialectical multi-agent, dynamic learner) $r = -0.087$, A2 cell_63 (dialectical multi-agent, bidirectional profiling) $r = -0.199$.

The single-turn–vs–multi-turn split has a clean pragmatic interpretation. In single-turn responses, ending with a question is the only available channel to cede initiative back to the learner; doing so signals "your turn now," which the rubric rewards as engaged tutoring. In multi-turn dialogues, the rubric scores the *full dialogue arc*; ending the *final* turn with a question means leaving the conversation unresolved, which the judges appear to penalise. The same surface feature signals different pragmatic acts depending on whether it occurs in a single-shot exchange or as the closing move of a multi-turn dialogue. (The §6.3 trajectory analyses already establish that turn-by-turn dynamics differ qualitatively from single-turn quality; the mediator's sign-reversal across this boundary is consistent with that broader pattern.)

The §7.10 mediator claim therefore has an explicit scope condition: **ends-with-question is a within-cell mediator in single-turn intersubjective cells (5 of 5 tested, r range +0.16 to +0.74, including replications across DeepSeek and Haiku ego, philosophy/programming/peer-support domains, and Sonnet/Opus/GPT judges); the relationship reverses sign in multi-turn intersubjective cells (4 of 5 tested, r range −0.09 to −0.30, with a single anomaly at cell_82 multi-turn base+superego, $r = +0.38$ at $n = 18$).** The cross-mode reversal does not falsify the underlying mechanism account (initiative-ceding); it specifies the discourse context in which the surface feature signals it. Report: `exports/d1-ends-question-replication.md`.

Neighbouring modules: Module 2 — Superego as Conditional Error Corrector; Module 3 — The Adaptive-Responsiveness Null; Module 10 — Model Dependence and Mechanism Boundary Conditions.

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